

Spend Now or Later? Dynamic Moral Hazard with a Fixed Budget

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September 21, 2026

Abstract

Organizations often give an agent a fixed budget to spend across time while relying on his effort to produce output. I develop a tractable model to characterize the optimal contract in a two-period moral-hazard problem with a fixed budget that can be used to reduce the cost of effort. I find that memory is still optimal in this setting. When the agent controls spending, he back-loads, concentrating resources in later periods where he expects to work the hardest. This resembles documented year-end spending spikes through incentives alone. When the principal controls spending, she front-loads instead, since winning early makes winning later more likely through memory. Delegating spending lowers the principal's profit, yet can raise total surplus when money is moderately effective at reducing effort cost.

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1 Introduction

Many organizations hand an agent a fixed budget to spend over time, and depend on that same agent’s effort to produce output. A division manager, for example, receives an annual capital budget from headquarters and decides how to allocate it across the year. The same constraint applies to a federal agency, which receives an annual budget from Congress, and to a Washington State agency, which receives a biennial budget from the state legislature. In each case, the agent decides how much of that fixed budget to spend now versus save for later within the same cycle, along with how much effort to exert in each period. This joint problem, intertemporal budget allocation under hidden effort, sits at the intersection of two literatures that otherwise stay separate. Capital budgeting mostly studies how budgets are allocated across projects or divisions under information frictions. This literature typically treats effort as observable or leaves it out of the model altogether. Dynamic moral hazard, on the other hand, studies hidden effort but typically without a fixed budget to allocate. This paper studies the joint problem in a tractable two-period model.

This joint problem is not simply of theoretical importance. Fixed-budget arrangements usually involve considerable sums, and organizations invest real effort in designing them.¹ A well-documented pattern in fixed-budget settings is that spending is concentrated toward the end of the cycle, as Figure 1 illustrates for U.S. federal contracts. Baumann (2019) find a similar pattern in U.K. government data, where roughly 16 percent of annual capital spending occurs in the final month of the fiscal year.² An increase in year-end spending is not, by itself, evidence that this pattern is wasteful. Late-cycle needs may simply be larger, as equipment may fail, or unresolved demands may accumulate over the year and only get addressed once other options are exhausted. The evidence, however, points to waste across both public and private settings. Liebman and Mahoney (2017) show that U.S. federal procurement spending in the last week of the fiscal year is 4.9 times the weekly average for the rest of the year, and that information-technology projects funded in that week receive substantially lower quality ratings.³ This pattern has been well studied, and

¹The U.S. federal government’s discretionary budget, the portion subject to annual appropriation, totaled roughly \$1.9 trillion in fiscal year 2025 (*Discretionary Spending in Fiscal Year 2025: An Infographic* 2026). Washington State’s operating budget for the 2025–27 biennium totals roughly \$80 billion; see (Washington Public Ports Association, 2026) for details.

²See Liebman and Mahoney (2017) for a similar rush documented in Canada, a pattern government officials there have called “March Madness”.

³Decaire and Sosyura (2023) document the same behavior among managers at publicly traded U.S. firms, who raise advertising spending sharply before budget-reset deadlines, with that spending yielding lower sales and margins.

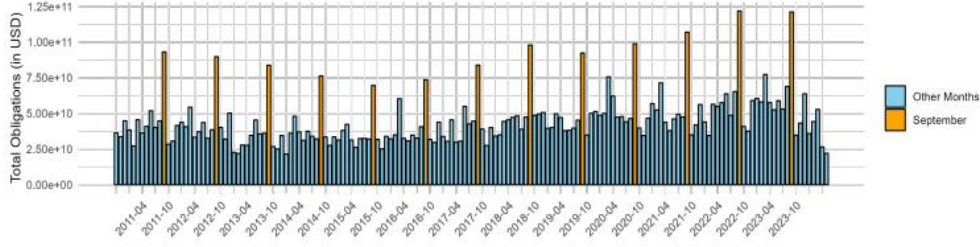


Figure 1: September (the final month of the U.S. fiscal year) contract spending is roughly twice the rest-of-year monthly average, consistently across 14 years. Source: Koller (2024), Figure 1; data from usaspending.gov.

existing work attributes it to mechanisms outside the contract itself. These explanations include use-it-or-lose-it rules, driven either by the agent’s motivation (Khalil et al., 2019) or by precautionary savings (Liebman and Mahoney, 2017). Other explanations point to the agent’s private utility from spending (Siemroth, 2024) or to procrastination (Baumann, 2019; Koller, 2024).

However, how the timing of spending within a cycle interacts with the effort of those who control it is largely unexplored. In the standard dynamic moral-hazard problem of Ohlendorf and Schmitz (2012), even though the two periods are technologically independent, the optimal contract still exhibits memory: second-period effort depends on the realized first-period outcome. A natural question is whether the memory property survives once a fixed budget is added to the same technologically independent setting, and how it interacts with the agent’s spending decision. A further question is whether the late-period spending spikes require any of the above mechanisms at all, or whether they can arise from dynamic incentives alone.

To that end, I add a fixed budget that can be allocated across periods to the standard two-period dynamic moral hazard setting. In particular, I consider a risk-neutral agent, protected by limited liability, who chooses both how much hidden effort to exert and how much of the fixed budget to spend in each period to reduce his effort cost. In this setting, money can buy complementary inputs that make effort cheaper without substituting for it.⁴ The probability of success is determined by the effort level alone. Spending itself carries no direct cost; the only friction is that the total budget is fixed, and spending more in one period leaves less available for the other.⁵ Thus, the trade-off in

⁴This assumption is discussed further in Section 2.

⁵This also means there is no underspending in the model by design. For a model in which underspending arises

this model is purely intertemporal. I also assume throughout that the principal can commit fully to the contract.⁶

I find that memory does, in fact, survive in this joint setting, and is a driver of the spending decision. The agent back-loads, that is, he spends more of the budget in the second period. A dollar of budget saves him more in cost where he expects to work hardest, which is the second period due to memory, so he holds resources back for that period. Effort, however, is not back-loaded the same way. The agent works exactly as hard as he is incentivized to in each period, but he would rather have money available when he expects to work hardest. The result resembles budget hoarding and the late-cycle spending patterns documented empirically, but provides a novel explanation for that pattern through only the dynamic structure of the contract itself.

If the principal instead had control of the spending while effort remained hidden, she would do the opposite and front-load. This, too, follows from memory, since a first-period success endogenously raises the probability of success later. As a result, the principal wants to make that first win as likely and as cheap to induce as possible. Early spending achieves this in two ways, by not only directly lowering the cost of inducing first-period effort, but also indirectly letting her promise a larger reward for that success in the second period. This discrepancy between the principal's choice and the agent's costs the principal profit. Therefore, it is a new cost of delegation which should be weighed against the other frictions delegation is usually meant to solve.

The budgets in question are often funded by taxpayers or shareholders, and the timing of spending has real consequences for the value that funding produces. This paper abstracts from the source of the money, but its allocation still has welfare consequences. To weigh these consequences, I consider a social planner who chooses how much to spend in each period to maximize total surplus. I find that the social planner front-loads even further than the principal does. This is partly because the principal counts rent paid to the agent as a cost, whereas the planner simply counts it as a transfer between two parties.

I compare total surplus across all three allocations. Revealed preference guarantees that the planner's allocation attains strictly higher welfare than the principal's. However, this guarantee does not extend to the agent's allocation, since the agent's control of spending is a different game

endogenously, see Koller (2024).

⁶Section 6 discusses renegotiation and how it might change the results of the paper.

altogether. In fact, delegating spending to the agent can raise total surplus when money is moderately effective at reducing effort cost. This gain does not come from the agent’s spending level, which actually costs surplus. Rather, to offset the agent’s back-loading, the principal offers a transfer schedule that pushes first-period effort closer to the social optimum. Essentially, there are two distortions at play: one in effort and one in spending. When money is only moderately effective at reducing cost, the effort distortion matters more for welfare, and delegation’s gain from correcting it dominates. As money becomes more effective, however, the spending distortion matters more instead. This, in turn, reverses the welfare gain from delegation.

The paper contributes to the literature on capital budgeting in firms. Harris and Raviv (1996) study capital allocation when division managers hold private information about project quality and have an incentive to overstate it. Harris and Raviv (1998) extend this to the choice between centralizing and delegating that allocation decision. The common friction in both papers is hidden information about project quality, and both consider a static setting. Johnson et al. (2017), on the other hand, extend it to a dynamic setting by studying a two-stage capital-budgeting problem in which managers across several divisions privately observe project information and compete for a scarce resource allocated through capital charge rates. All three, however, abstract away from any effort choice by the agent. Rogerson (1997) does include an effort choice, and designs depreciation rules that align a manager’s investment timing with net present value, but that effort is observable. Bernardo et al. (2001) comes closest to this paper’s setting, allowing both hidden project quality and hidden effort. They do not consider a fixed budget, however, so the agent faces no intertemporal allocation problem.

The paper also relates to the literature on dynamic moral hazard. Memory in dynamic optimal contracts is a well-established result. Rogerson (1985), Lambert (1983), and Townsend (1982) establish it in settings with risk-averse agents, where the mechanism is consumption-smoothing. Ohlendorf and Schmitz (2012) show that the property extends to a setting with a risk-neutral agent protected by limited liability. None of these papers, however, considers any second hidden action alongside effort. I adopt the risk-neutral setting and add a fixed budget, which the agent allocates across the two periods and is hidden from the principal. Taken together, the capital budgeting and dynamic moral hazard literatures are extensive, but combining them has remained largely unexplored.

Back-loading is also specifically documented and explained in the literature on year-end government and corporate spending. Liebman and Mahoney (2017) model it as precautionary saving. Agencies hold funds back for uncertain future needs, and when those needs fail to materialize, agents spend the remainder anyway under use-it-or-lose-it rules. Baumann (2019) argue for procrastination: the agency discounts the value of production the further it is from the fiscal year-end, which makes early spending relatively unattractive. Koller (2024) combine both mechanisms, precautionary saving and procrastination, to generate the spending spike in his model. Siemroth (2024) instead gives the agent private information about the project’s spending needs and lets him derive private utility from spending above what the project actually requires. None of these four models considers an effort choice for the agent. Khalil et al. (2019) includes a hidden effort choice for the agent, but they explain spending spikes through the agent’s misalignment with the principal’s mission. A more misaligned agent spends less on productive uses throughout the year and is then forced to spend the leftover under use-it-or-lose-it. In their model, output is unobservable and not contractible, so the budget rule is the only available incentive instrument.

A related strand of the principal-agent literature has the agent exert hidden effort while the principal controls a second resource. Ishiguro and Yasuda (2018) study multi-agent moral hazard in which the principal fixes a total wage budget across agents, independent of realized output. Bond and Gomes (2009) consider multi-task settings in which the agent’s allocation of effort across tasks is hidden, while the principal chooses how much weight to assign to each task. Goel and Hann-Caruthers (2024) characterize when weighted bonus contracts are optimal in a multidimensional environment with a fixed budget for rewards. Khalil et al. (2013) study a three-level hierarchy in which a politician delegates contract design to a bureaucrat, who then sets both the agent’s contract and the budget allocation in a static setting. In each case, the agent has no decision to make over how the second resource, budget or otherwise, is used.

In this paper, however, the agent controls both hidden actions. Multidimensional moral hazard problems with non-separable costs are rarely tractable. Therefore, an additional contribution of this paper is a tractable model in which two hidden actions jointly determine the agent’s cost. Holmstrom and Milgrom (1991) showed that with several hidden actions jointly determining output, the incentive constraints become entangled. Further, **fernandes2000nonpersistent** show that the problem worsens in dynamic settings, since a second hidden action typically has to be tracked as

an additional state variable. This model avoids both problems. Spending changes only the cost of effort, so the incentive constraint for effort does not depend on spending. The budget path also does not need to be separately tracked. The agent commits to first-period spending before any outcome is realized. The second period is terminal, and since money is costless, there is no longer an intertemporal trade-off. The agent spends the remainder of the budget.

The rest of the paper is organized as follows. Section 2 sets up the two-period model and the interpretation of spending as a cost-reducer for effort. Section 3 provides the static benchmarks and the optimal contracts in the dynamic setting. Section 4 introduces the social-planner benchmark and compares welfare across regimes. Section 5 considers discounting and how the results depend on how effective spending is at reducing effort cost. Finally, Section 6 concludes.

2 Model

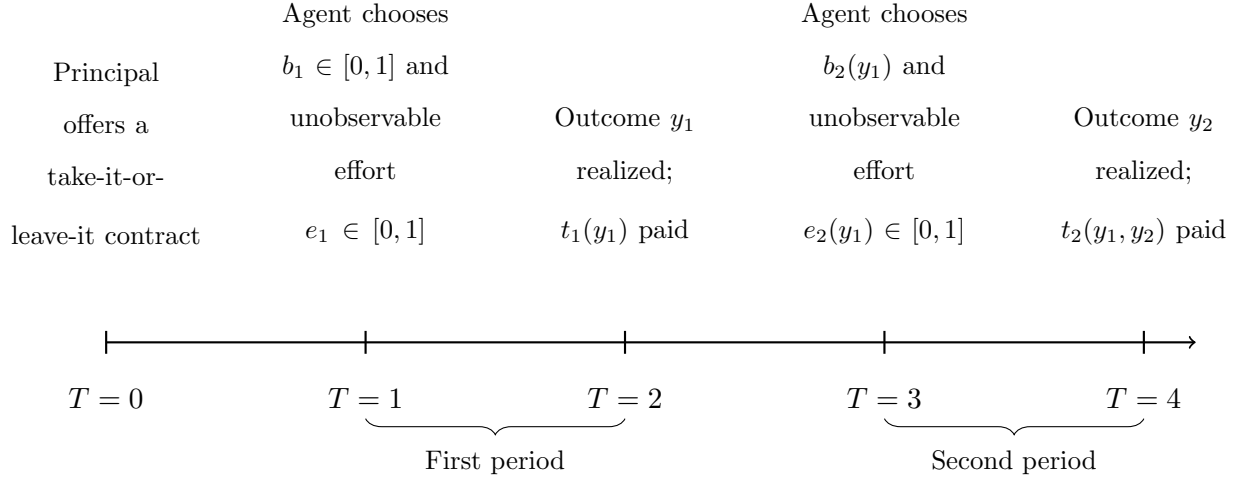
The paper adapts the dynamic moral-hazard model of Ohlendorf and Schmitz (2012), adding a fixed budget that can be allocated across the two periods. There are two parties, a principal (she) and an agent (he), both of whom are risk neutral. The agent is protected by limited liability arising from liquidity constraints, so transfers must be non-negative. The parties' reservation utilities are zero. At the initial date $T = 0$, the principal makes a take-it-or-leave-it offer to the agent for two periods. If the agent rejects, the game ends and both receive their reservation utility.

Upon accepting, the agent exerts unobservable first-period effort $e_1 \in [0, 1]$ at date $T = 1$ and spends $b_1 \in [0, 1]$ from a fixed total budget $B = 1$, incurring a cost of effort $c(e_1, b_1)$. At date $T = 2$, either a success ($y_1 = 1$) or a failure ($y_1 = 0$) is realized, with the probability of success normalized to equal the effort level, $\Pr(y_1 = 1 \mid e_1) = e_1$. The principal's verifiable return is the outcome y_1 , and the transfer $t_1(y_1) \geq 0$ is paid to the agent. The agent then chooses unobservable second-period effort $e_2(y_1) \in [0, 1]$ at date $T = 3$ and spends $b_2(y_1) \in [0, 1 - b_1]$ from the remaining budget, incurring a cost of effort $c(e_2(y_1), b_2(y_1))$.⁷ Finally, at date $T = 4$, the verifiable second-period return $y_2 \in \{0, 1\}$ is realized, with $\Pr(y_2 = 1 \mid e_2(y_1)) = e_2(y_1)$, and the transfer $t_2(y_1, y_2) \geq 0$ is paid to the agent.⁸

⁷I assume the agent cannot be replaced at date $T = 2$ or later, for instance because hiring a new agent would require costly project-specific retraining.

⁸The constraint $t_2(y_1, y_2) \geq 0$, rather than the weaker cumulative form $t_2(y_1, y_2) \geq -t_1(y_1)$, reflects that the agent cannot be forced to return pay already received.

The two periods are technologically independent in the sense that a first-period success or failure does not make second-period success or failure more probable: y_2 depends only on e_2 , not on e_1 or y_1 . This rules out persistence in productivity and isolates the dynamic mechanism of interest, which operates through the contract rather than through technology. I assume the principal can commit at $T = 0$ not to renegotiate the contract.⁹ The timing of events is illustrated below:



and that tradeoff is simple: the budget can be spent now, to make first-period effort cheaper, or saved, to make second-period effort cheaper.

This is a deliberate modeling choice, for two reasons. First, it keeps the model tractable.¹¹ Second, it isolates incentives as the force that drives the timing of spending. If money were productive, spending would buy success directly, and its timing would partly reflect that mechanical effect rather than incentives. With money confined to the cost side, spending affects payoffs only through the cost of effort, so the timing of spending across periods responds to the dynamic incentives over effort rather than to any direct effect on output.

Substantively, the budget can stand for discretionary inputs that make effort cheaper rather than generating output on their own. For instance, a funding authority might allocate a fixed appropriation to a department that spends it on support staff and analytical tools, while the quality of policy outcomes still depends on the department’s own effort; or a firm might give a sales manager a fixed expense budget for travel and client support, while closing deals still depends on the manager’s effort. In each case the budget makes effort cheaper without substituting for it.

Remark 1 (Periods as phases of the budget cycle). The two periods of the model are best read as two *phases* of a single fixed-budget cycle (e.g., the first and second halves of a fiscal year), not as two instants in time. This interpretation is consistent with how Liebman and Mahoney (2017) model within-cycle spending, with agencies building up funds early and drawing them down toward the close of the cycle.

3 Optimal Contracting

Notation. The subscript S denotes the static (one-period) problem and the subscript D the dynamic (two-period) problem; in the dynamic problem a period numeral distinguishes first- from second-period objects, as in e_{1D} and $e_{2D}(y_1)$. Superscripts denote the contracting regime: FB for the first best, SB for the static second best, P for principal-controls, and A for agent-controls.

effort choice interior and the effort first-order conditions valid.

¹¹If money entered the output technology, the agent would control two hidden actions, effort and spending, both bearing on a single output signal. This is a multitask moral-hazard problem, in which the incentive constraints entangle and the first-order approach need not be valid (Holmstrom and Milgrom, 1991; Rogerson, 1985; Fernandes and Phelan, 2000). The cost-side formulation keeps the effort incentive constraints one-dimensional, a point I return to in Section 3.4.

Expected profit to the principal Π_D and agent rent A_D summarize the entire dynamic problem and carry no period numeral.

3.1 Static benchmark

I begin with a one-period version of the model. The static benchmark serves two purposes: it pins down the moral-hazard wedge that the dynamic results build on, and it shows that hidden b creates no distortion when the contracting horizon collapses to a single period. The budget conflict between principal and agent is therefore a purely dynamic phenomenon.

I work through three contracting environments that mirror the dynamic analysis: a first-best with both actions observable, a principal controls spending case with hidden effort, and an agent controls spending case with both effort and spending hidden. Under agent controls spending, two further subcases arise in which only one action is hidden. Both collapse onto an outcome already covered by the three main cases: observing effort recovers the first-best, while observing spending generates the second-best outcome of the principal-controls case.¹²

In every static case the principal and the agent agree on $b = 1$, that is, fully spending the available budget, since it is costless to spend and worthless if left unspent. No party deviates from that common choice, so it makes no difference whether b is observed or hidden, and the outcome turns only on whether effort is observable.

3.1.1 First-best

The principal maximizes expected profit, the probability of success net of the expected transfer paid to the agent. With both effort and spending observable the IR constraint binds, so the transfer covers the agent's effort cost exactly and leaves no rent. The principal's problem simplifies to

$$\max_{e \in [0,1], b \in [0,1]} e - \frac{e^2}{(1+b)^n}. \quad (2)$$

Once effort is optimized out, the principal's objective depends on b alone and is strictly increasing in spending, so the principal exhausts the budget. Money lowers the cost of effort and is privately

¹²Neither subcase adds a new result, so both are omitted here; the derivations are provided in Appendix A.

costless, which is why more of it always helps. At the optimum,

$$b_S^{\text{FB}} = 1, \quad e_S^{\text{FB}} = \frac{2^n}{2}, \quad \Pi_S^{\text{FB}} = \frac{2^n}{4}, \quad A_S^{\text{FB}} = 0.$$

3.1.2 Principal controls spending, effort unobservable

With effort hidden, the principal posts a transfer schedule $t : \{0, 1\} \rightarrow \mathbb{R}_+$ and the agent chooses e to maximize his expected transfer net of effort cost,

$$e t(1) + (1 - e) t(0) - \frac{e^2}{(1 + b)^n}. \quad (3)$$

Limited liability binds the failure transfer at $t(0) = 0$, and the agent's effort incentive constraint ties the success transfer $t(1)$ to the effort it induces. With the transfer substituted into expected profit, the principal's problem simplifies to

$$\max_{e \in [0, 1], b \in [0, 1]} e - \frac{2e^2}{(1 + b)^n}. \quad (4)$$

The induced effort is half the first-best level at the same budget. Once effort is optimized out, the objective in b alone is again strictly increasing, so the principal fully spends the budget. At the optimum,

$$b_S^P = 1, \quad e_S^P = \frac{2^n}{4}, \quad \Pi_S^P = \frac{2^n}{8}.$$

To make the rent precise, I define

$$A(e) := e c'(e) - c(e), \quad (5)$$

the agent's rent from a contract that induces effort e , where $c'(e) = c_e(e, b)$ is the marginal effort cost at the prevailing budget.¹³ The rent is non-negative and strictly increasing, with $A'(e) = e c''(e)$, so higher induced effort commands a higher rent. The principal holds effort below the first-best level to economize on that rent, resulting in the standard moral-hazard distortion. She cannot eliminate it, though, and at the optimum the agent earns a positive rent, $A_S^P = \frac{2^n}{16}$. Spending is

¹³The rent depends on the budget through the cost function, but the budget is held fixed wherever A is used.

again privately costless, so she spends up to the budget limit.

3.1.3 Agent controls spending, both effort and spending unobservable

With both effort and spending hidden, transfers again depend only on output, $t : \{0, 1\} \rightarrow \mathbb{R}_+$, and the agent now chooses how much to spend, b , as well as how much effort to exert, e . His effort objective is the same as in the principal-controls case, given by (3). For any nonnegative output-contingent transfer his payoff is increasing in b , so he spends the entire budget, $b_S^{\text{SB}} = 1$. Anticipating this, the principal solves the same hidden-effort problem as before, and the induced outcome is identical,

$$e_S^{\text{SB}} = \frac{2^n}{4}, \quad \Pi_S^{\text{SB}} = \frac{2^n}{8}, \quad A_S^{\text{SB}} = \frac{2^n}{16}.$$

With both actions hidden, a single transfer schedule induces a pair (e, b) rather than effort alone. This does not complicate the problem here, since $b = 1$ is optimal for the agent regardless of the schedule. Spending only lowers his cost, so the budget choice is pinned down by technology rather than by the contract, and the principal's problem reduces to the standard hidden-effort one.

The static outcome is robust to the observability of b because the principal and the agent are aligned on the budget choice. The principal wants to exhaust the budget because it lowers the cost of inducing effort; the agent wants the same because spending lowers his own cost of effort and is privately free. The two subcases in Appendix A confirm this, since hiding b alone or hiding e alone leaves the equilibrium outcome unchanged.

Proposition 1 (Static benchmark). *In the static one-shot game, the principal and the agent both choose $b = 1$ regardless of who controls spending and of what is observable. The equilibrium outcome depends only on whether effort is observable, and takes two forms:*

(i) First-best (when effort is observable): $e_S^{\text{FB}} = \frac{2^n}{2}$, $\Pi_S^{\text{FB}} = \frac{2^n}{4}$, $A_S^{\text{FB}} = 0$.

(ii) Second-best (when effort is hidden): $e_S^{\text{SB}} = \frac{2^n}{4}$, $\Pi_S^{\text{SB}} = \frac{2^n}{8}$, $A_S^{\text{SB}} = \frac{2^n}{16}$.

Hidden b creates no additional distortion beyond the standard moral-hazard wedge from hidden effort.

The proof is provided in Appendix A. Principal and agent are aligned when the horizon is a single period, so any non-trivial budget-allocation distortion must come from dynamic incentives. The

dynamic case introduces an inter-period link in the agent's first-period IC, and through it hidden spending begins to matter. The remainder of Section 3 characterizes the optimal contracts in the dynamic setting.

3.2 Dynamic first-best

I now turn to the two-period model. The dynamic first best serves as the natural benchmark for the dynamic results that follow. With both effort and spending observable, the IR binds and the principal chooses $(e_1, e_2(\cdot), b_1, b_2(\cdot))$ to maximize total expected output net of effort costs,

$$\max e_1 \left(1 + e_2(1) - \frac{e_2(1)^2}{(1 + b_2(1))^n} \right) + (1 - e_1) \left(e_2(0) - \frac{e_2(0)^2}{(1 + b_2(0))^n} \right) - \frac{e_1^2}{(1 + b_1)^n}, \quad (6)$$

subject to the budget constraint $b_2(y_1) \leq 1 - b_1$. The cost is strictly decreasing in spending for any positive effort, and left-over budget has no continuation value, so the principal exhausts the second-period budget at every history,

$$b_2(y_1) = 1 - b_1, \quad y_1 \in \{0, 1\}. \quad (7)$$

The non-trivial budget-allocation question is therefore the choice of b_1 , which fixes the second-period budget through (7).

With effort observable, there is no reason to make second-period effort respond to the first-period outcome. Therefore, the first-best implements the same second-period effort after either first-period outcome,

$$e_1^{\text{FB}} = \frac{(1 + b_1)^n}{2}, \quad e_2^{\text{FB}}(y_1) = \frac{(2 - b_1)^n}{2}, \quad y_1 \in \{0, 1\}. \quad (8)$$

The budget split then trades off cheaper first-period effort from a marginal unit of first-period spending against the more expensive second-period effort that a smaller residual budget implies. The way this resolves depends entirely on the curvature parameter n , which governs how strongly a marginal unit of spending lowers effort cost. Three cases arise.

Decreasing returns ($n < 1$).

The principal's objective is concave in spending, so the optimum is interior and, by symmetry, an even split, $b_1^{\text{FB}} = \frac{1}{2}$.

Constant returns ($n = 1$).

The objective is constant in spending, so any allocation is optimal, $b_1^{\text{FB}} \in [0, 1]$.

Increasing returns ($n > 1$).

The objective is convex in spending, so the optimum is a corner, $b_1^{\text{FB}} \in \{0, 1\}$.

The economic logic is straightforward. Under decreasing returns, spending within a period has diminishing marginal value. A unit of budget is worth more where less has been spent, so the principal smooths spending across the two periods. Under increasing returns, the marginal value of spending rises, and concentrating the whole budget in one period beats splitting it. At constant returns the two periods are perfect substitutes, so the split is indeterminate.

I focus on the decreasing-returns case where $n < 1$ for the rest of the paper. In this regime, the principal splits the budget evenly across the two periods under full information. This provides a clean symmetric benchmark against which to measure the budget allocation distortions that arise once effort is unobservable. In particular, I set $n = \frac{1}{2}$ for ease of exposition.¹⁴ With spending split evenly, $b_1 = \frac{1}{2}$, both effort functions collapse to the same constant across periods and histories,

$$e_{1D}^{\text{FB}} = e_{2D}^{\text{FB}}(1) = e_{2D}^{\text{FB}}(0) = \frac{\sqrt{6}}{4},$$

and the principal's profit equals total surplus,

$$\Pi_{1D}^{\text{FB}} = \frac{\sqrt{6}}{4}.$$

The proposition below states the dynamic first-best contract, with the proof in Appendix A.

Proposition 2 (Dynamic first-best). *With effort and spending both observable, the principal exhausts the second-period budget at every history, $b_2(y_1) = 1 - b_1$, and implements identical second-*

¹⁴The qualitative budget allocation results hold across the relevant range $n \in (0, 1)$, though the welfare comparison carries a caveat that I discuss in Section 5.1.

period effort across histories,

$$e_1^{\text{FB}} = \frac{(1 + b_1)^n}{2}, \quad e_2^{\text{FB}}(y_1) = \frac{(2 - b_1)^n}{2}.$$

Fixing $n = \frac{1}{2}$, the optimal contract implements

$$b_1^{\text{FB}} = \frac{1}{2}, \quad b_2^{\text{FB}}(y_1) = \frac{1}{2}, \quad e_{1D}^{\text{FB}} = e_{2D}^{\text{FB}}(y_1) = \frac{\sqrt{6}}{4} \approx 0.612, \quad \Pi_{1D}^{\text{FB}} = \frac{\sqrt{6}}{4} \approx 0.612, \quad A_{1D}^{\text{FB}} = 0.$$

3.3 Recap: Standard dynamic moral hazard optimal long-term contract with $B = 0$

Before turning to the cases of interest, where effort is hidden and spending decisions must be made, I describe the optimal contract when there is no budget to allocate. This recap of the standard case will provide the foundation for understanding the results once budget and spending enter the model. Setting $B = 0$ removes spending from the model and leaves the standard two-period moral hazard problem of Ohlendorf and Schmitz (2012). In this setup, effort remains hidden in both periods, but the agent has no spending decision to make in either one.

The optimal long-term contract in this standard case displays *memory*, that is, second-period effort is conditional on the first-period outcome. Specifically, the agent is induced to exert higher second-period effort after a first-period success than after a first-period failure.¹⁵ This gives rise to an endogenous hot hand effect: succeeding early makes succeeding in the second period more likely, without any technological spillover between periods.

Memory makes providing first-period incentives cheaper, because the second period can now be used to incentivize as well. The principal can motivate the agent in two ways: she can pay the agent directly for an early success, or she can promise a larger continuation rent on the good branch than on the bad. She does the latter by inducing higher effort on the good branch. Since rent increases with effort, this difference in effort creates a wedge between the two branches' continuation rents, which the principal draws on to provide first-period incentives at a lower cost.

¹⁵I will often refer to the branch following a first-period success ($y_1 = 1$) as the *good* branch, and the branch following a first-period failure ($y_1 = 0$) as the *bad* branch.

3.4 Agent controls spending

The agent chooses both effort and spending in each period, and neither is observable to the principal, so the contract specifies only the transfer schedule $(t_1(y_1), t_2(y_1, y_2))$. I solve the problem backward, from the second period to the first, and then characterize the optimal contract.

Second-period contract. For a fixed first-period budget b_1 and history y_1 , the agent jointly chooses $(e_2(y_1), b_2(y_1))$ to maximize

$$e_2(y_1) t_2(y_1, 1) + (1 - e_2(y_1)) t_2(y_1, 0) - \frac{e_2(y_1)^2}{(1 + b_2(y_1))^n},$$

for non-negative $t_2(y_1, 1)$ and $t_2(y_1, 0)$, due to limited liability. The agent's second-period effort incentive constraint is characterized by the first-order condition,¹⁶

$$e_2(y_1) = \frac{(1 + b_2(y_1))^n}{2} [t_2(y_1, 1) - t_2(y_1, 0)]. \quad (\text{IC}_2(y_1))$$

The effort cost is strictly decreasing in b for $e_2(y_1) > 0$, and since the second period is terminal, any unspent budget carries no value. The agent therefore exhausts the second-period budget at every history,

$$b_2(y_1) = 1 - b_1, \quad y_1 \in \{0, 1\}. \quad (9)$$

First-period effort IC. The agent's continuation rent after history y_1 is

$$a(y_1) = t_1(y_1) + t_2(y_1, 0) + \frac{e_2(y_1)^2}{(2 - b_1)^n}, \quad (10)$$

where I have substituted $(\text{IC}_2(y_1))$ at the binding budget. The agent's first-period effort solves

$$\max_{e_1} e_1 a(1) + (1 - e_1) a(0) - \frac{e_1^2}{(1 + b_1)^n},$$

¹⁶The first-order condition is sufficient because the agent's objective is strictly concave in effort. The expected transfer is linear in effort and the effort cost is convex, so the condition pins down the global optimum. The same holds for the first-period effort choice below.

with first-period effort incentive constraint

$$a(1) - a(0) = \frac{2e_1}{(1+b_1)^n}. \quad (11)$$

Substituting the continuation rents from (10) gives the explicit form

$$\frac{2e_1}{(1+b_1)^n} = [t_1(1) - t_1(0)] + [t_2(1,0) - t_2(0,0)] + \frac{e_2(1)^2 - e_2(0)^2}{(2-b_1)^n}. \quad (\text{IC}_1)$$

First-period incentives come from the spread in the first-period transfers, the spread in second-period failure transfers, and the second-period rent wedge.¹⁷

First-period budget IC. The agent's first-period choice of b_1 adds a constraint absent from the standard dynamic moral hazard model. Let

$$V(b_1) = \max_{e_1} U(e_1, b_1)$$

be his value from optimizing effort at a given b_1 . The budget incentive constraint is

$$b_1 \in \arg \max_{b_1 \in [0,1]} V(b_1).$$

Unlike the effort choice, $V(b_1)$ need not be concave in b_1 , so the first-order condition is necessary but not sufficient. I solve it numerically and confirm the solution is a maximum by evaluating $V(b_1)$ over a grid of b_1 .¹⁸ By the envelope theorem at the optimal $e_1^*(b_1)$, and using (10), the derivative is

$$\frac{dV}{db_1} = \frac{ne_1^2}{(1+b_1)^{n+1}} - \frac{n[e_1 e_2(1)^2 + (1-e_1)e_2(0)^2]}{(2-b_1)^{n+1}} = 0. \quad (\text{IC}_{b_1})$$

The condition balances the marginal benefit of first-period spending against its marginal cost. The first term is the benefit, which captures the saving on first-period effort cost. The second term is the cost, which is the forgone saving on second-period effort cost, averaged over the two branches by their probabilities. In short, spending more today lowers the cost of effort now but raises it

¹⁷I discuss this equation once the principal's problem pins down the failure transfers at zero, and only the direct transfer and rent wedge survive in determining first-period effort.

¹⁸Numerical implementation and verification are provided in Appendix B.

later.

Principal's problem. The principal chooses the transfer schedule $(t_1(y_1), t_2(y_1, y_2))$ to maximize expected profit, subject to $(\text{IC}_2(y_1))$, (IC_1) , (IC_{b_1}) , and limited liability. Limited liability requires every transfer to be non-negative. As the effort incentive constraints above show, effort responds only to the spread between success and failure transfers. Lowering a failure transfer, together with its paired success transfer by the same amount to preserve the spread, therefore strictly lowers expected payment without changing induced effort. The principal drives every failure transfer to zero, where limited liability binds,

$$t_1(0) = t_2(1, 0) = t_2(0, 0) = 0.$$

The first-period incentive constraint (IC_1) then reduces to

$$\frac{2e_1}{(1+b_1)^n} = t_1(1) + \frac{e_2(1)^2 - e_2(0)^2}{(2-b_1)^n}. \quad (12)$$

This is the central equation of the dynamic problem. The first term $t_1(1)$ provides direct first-period incentives. The second term is the second-period rent wedge, which provides indirect incentives by making the good branch more rewarding than the bad. Widening the wedge, by inducing higher $e_2(1)$ or lower $e_2(0)$, reduces the direct transfer $t_1(1)$ the principal must pay to induce a given e_1 . This is the same mechanism present in the standard case in Section 3.3: paying directly for first-period effort and rewarding it indirectly through a wider second-period rent wedge are substitute instruments.

The principal's problem reduces to a choice of the three success transfers $(t_1(1), t_2(1, 1), t_2(0, 1))$, which yield expected profit

$$\Pi = e_1(1 + e_2(1)(1 - t_2(1, 1)) - t_1(1)) + (1 - e_1)e_2(0)(1 - t_2(0, 1)).$$

The effort ICs express e_1 and $e_2(\cdot)$ in the transfers, and the budget IC pins down b_1 as the interior root of (IC_{b_1}) .¹⁹ This root has no closed form in the transfers, so b_1 cannot be substituted out and

¹⁹This root is unique in every case I compute; see Appendix B.

profit is not available as an explicit function of the transfer triple. I solve the problem numerically.

Proposition 3 (Agent controls spending contract). *At $n = \frac{1}{2}$ the optimal agent controls spending contract yields*

$$b_1^A \approx 0.33, \quad e_{1D}^A \approx 0.34, \quad e_{2D}^A(1) \approx 0.59, \quad e_{2D}^A(0) \approx 0.25,$$

with profit $\Pi_{1D}^A \approx 0.333$ and agent rent $A_{1D}^A \approx 0.146$. The transfers implementing the outcome are $t_1(1) \approx 0.36$, $t_2(1, 1) \approx 0.92$, and $t_2(0, 1) \approx 0.39$.

The derivation is in Appendix A.3; the numerical procedure used to solve for the contract is in Appendix B. The contract has two features. First, it displays memory, that is, second-period effort depends on the first-period outcome, as can be seen by $e_{2D}^A(1) > e_{2D}^A(0)$. Second, the agent *back-loads* the budget, spending less than half in the first period, $b_1^A < \frac{1}{2} = b_1^{\text{FB}}$. The induced effort is ordered as follows:

$$e^{\text{FB}} > e_{2D}^A(1) > e_{1D}^A > e_{2D}^A(0).$$

Memory follows from the effort incentive constraints as in the standard problem of Section 3.3. If the principal induced no memory instead, setting $t_1(1) = t_2(1, 1) = t_2(0, 1)$, the agent would split the budget evenly, $b_1 = \frac{1}{2}$.²⁰ The principal still finds memory optimal, since it remains a cheaper way to provide first-period incentives, and this is exactly what breaks the symmetry in spending.

Back-loading, the agent spending more in the second period, is a by-product of memory, and it traces to the budget IC (IC_{b_1}). Money is a cost-reducer (Section 2), and the marginal cost-saving from a unit of budget rises with the effort level induced. This creates an intertemporal trade-off: spending now makes reaching the good branch cheaper, while holding budget back lets the agent exploit the good branch once he is there. The agent prefers to save and hold budget for the branch where expected effort is highest, since that is where a unit of budget is worth the most.

Evaluating the budget FOC at $b_1 = \frac{1}{2}$ makes the argument more precise. The sign of the derivative reduces to the sign of $E[e_2^2] - e_1^2$.²¹ When

$$E[e_2^2] > e_1^2, \quad E[e_2^2] = e_1 e_2(1)^2 + (1 - e_1) e_2(0)^2,$$

²⁰This would implement the first-best allocation with respect to spending, though not with respect to effort.

²¹The agent is risk neutral over money; the second moment enters through the convexity of the cost function, not through risk aversion.

the agent gains by shifting budget toward the second period, placing the optimal first-period budget below $\frac{1}{2}$. Substantively, the condition says that expected second-period effort, and so the expected cost-saving from a unit of second-period budget, is higher. This condition holds in the optimal contract, so the agent back-loads.

This gap between $E[e_2^2]$ and e_1^2 has two sources. First, memory itself is a source of volatility, as second-period effort, unlike first-period effort, takes one of two values depending on the branch. Since effort cost is convex, this volatility alone raises $E[e_2^2]$. Second, memory raises the average as well, since $e_2(1)$ is induced above $e_2(0)$ to reward first-period success. Both effects push $E[e_2^2]$ above e_1^2 , so budget is worth more in the second period than the first. Effort itself is not back-loaded, however. The agent works hard whenever incentivized to, but saves money for the period where a unit of budget delivers the largest cost-saving.

The pattern is qualitatively consistent with budget hoarding and end-of-cycle spending spikes documented in the year-end spending literature, though the mechanism here is generated by dynamic incentives rather than by use-it-or-lose-it rules, procrastination (Koller, 2024; Baumann, 2019), or private utility from spending (Siemroth, 2024). Under use-it-or-lose-it rules, the spike can arise either from precautionary saving (Hurley et al., 2014; Liebman and Mahoney, 2017) or from the agency’s own motivation (Khalil et al., 2019).^{22,23}

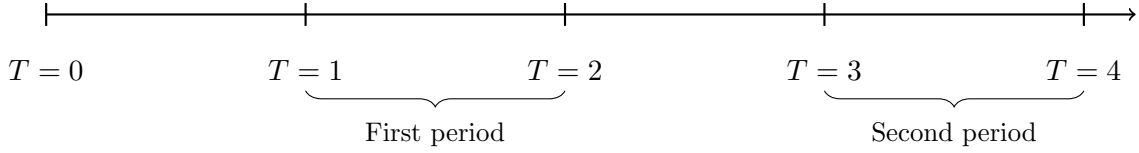
3.5 Principal controls spending, agent exerts unobservable effort

I now turn to the case where effort is private to the agent in both periods, but the principal allocates the budget herself. The timing is illustrated below:

²²The precautionary saving channel is the closest of these, in which the agent also saves resources against a state where a unit of budget delivers the largest value. In that setting, the state is determined by exogenous demand realized late in the cycle. Year-end spikes occur when the anticipated need fails to materialize, forcing low-quality spending before funds expire. In this model, the source of the state is the good branch generated by the contract itself, and back-loading results without any demand uncertainty.

²³Under the period-as-phase reading of Remark 1, $b_1^A < \frac{1}{2}$ is comparable to empirical measures such as the share of capital spending in the final month of Baumann (2019) or the last-week-to-average-week procurement ratio of Liebman and Mahoney (2017).

	Principal sets			
	$b_1 \in [0, 1];$			
Principal	agent chooses		Agent chooses	
offers a	unobservable	Outcome y_1	unobservable	Outcome y_2
take-it-or-	effort	realized;	effort	realized;
leave-it contract	$e_1 \in [0, 1]$	$t_1(y_1)$ paid	$e_2(y_1) \in [0, 1]$	$t_2(y_1, y_2)$ paid



Substituting these into expected profit and simplifying gives the reduced-form objective

$$\begin{aligned}\Pi(e_1, e_2(0), e_2(1), b_1) = & e_1 \left(1 + e_2(1) - \frac{e_2(1)^2}{(2-b_1)^n} \right) - \frac{2e_1^2}{(1+b_1)^n} \\ & + e_2(0) - \frac{2e_2(0)^2}{(2-b_1)^n} - e_1 \left(e_2(0) - \frac{e_2(0)^2}{(2-b_1)^n} \right).\end{aligned}\quad (13)$$

The principal chooses $(e_1, e_2(0), e_2(1), b_1)$ to maximize (13) directly. The first-order conditions are as follows. The $[e_2(1)]$ -FOC gives the closed form

$$e_2(1) = \frac{(2-b_1)^n}{2}.\quad (14)$$

The $[e_1]$ -FOC, after substituting (14), and the $[e_2(0)]$ -FOC are

$$\begin{aligned}[e_1]: \quad & 1 + \frac{(2-b_1)^n}{4} - \frac{4e_1}{(1+b_1)^n} - e_2(0) + \frac{e_2(0)^2}{(2-b_1)^n} = 0, \\ [e_2(0)]: \quad & 1 - \frac{4e_2(0)}{(2-b_1)^n} - e_1 + \frac{2e_1e_2(0)}{(2-b_1)^n} = 0.\end{aligned}$$

These two equations, together with (14), determine the effort levels at any fixed b_1 . They are the same equations that pin down effort in Section 3.4 once b_1 is held fixed. Since b_1 is a choice variable for the principal, the first-order condition with respect to b_1 is given by

$$\frac{2ne_1^2}{(1+b_1)^{n+1}} + \frac{ne_1e_2(1)^2}{(2-b_1)^{n+1}} = \frac{n(2e_1e_2(1)^2 + 2e_2(0)^2 - e_1e_2(0)^2)}{(2-b_1)^{n+1}}.\quad (15)$$

The left-hand side is the marginal benefit of raising b_1 , and the right-hand side is its marginal cost. With $e_2(1)$ given by (14), the $[e_1]$, $[e_2(0)]$, and $[b_1]$ -FOCs form a system of three equations, nonlinear in $(e_1, e_2(0), b_1)$, with no closed-form solutions for e_1 and b_1 . The $[e_2(0)]$ -FOC does, however, solve explicitly for $e_2(0)$ in terms of e_1 and b_1 (Appendix A.4), so the numerical problem at each b_1 reduces to a single equation in e_1 . I solve for the optimal contract numerically.

Proposition 4 (Principal controls spending contract). *At $n = \frac{1}{2}$ the optimal principal controls spending contract yields*

$$b_1^P \approx 0.53, \quad e_{1D}^P \approx 0.34, \quad e_{2D}^P(1) \approx 0.61, \quad e_{2D}^P(0) \approx 0.24,$$

with profit $\Pi_{1D}^P \approx 0.336$ and agent rent $A_{1D}^P \approx 0.143$. The transfers implementing the outcome are $t_1(1) \approx 0.30$, $t_2(1, 1) \approx 1.00$, and $t_2(0, 1) \approx 0.40$.

The derivation is provided in Appendix A.4 and details about the numerical procedure are in Appendix B. Unlike the agent, the principal front-loads the budget, spending more in the first period. To see why, split each side of (15) into two terms,

$$MC = \underbrace{\frac{n[2e_1 e_2(1)^2 + 2(1 - e_1) e_2(0)^2]}{(2 - b_1)^{n+1}}}_{(i)} + \underbrace{\frac{n e_1 e_2(0)^2}{(2 - b_1)^{n+1}}}_{(ii)}, \quad (16)$$

$$MB = \underbrace{\frac{2n e_1^2}{(1 + b_1)^{n+1}}}_{(iii)} + \underbrace{\frac{n e_1 e_2(1)^2}{(2 - b_1)^{n+1}}}_{(iv)}. \quad (17)$$

Term (i) is the back-loading force, which represents part of the marginal cost of raising spending in the first period. This is because raising b_1 shrinks the second-period budget, raising the transfers needed to sustain effort on both branches there. Term (iii) is the front-loading force, as raising b_1 directly lowers the cost of inducing e_1 .²⁴

Terms (ii) and (iv) form the indirect channel and arise from the rent wedge. Raising b_1 raises the continuation rent in both branches. On the failure branch, term (ii) represents the narrowing of the rent wedge, which raises the direct transfer the principal must pay in the first period. On the success branch, term (iv) represents the widening of the wedge instead. The memory property means that the principal finds it optimal to induce higher second-period effort following a first-period success compared to a failure, which implies that the good branch's rent rises by more.²⁵ Thus, term (iv) exceeds term (ii) and the wedge widens on net from additional spending in the first period. Memory, therefore, raises the marginal benefit of an additional unit of first-period spending, and this is what pushes b_1^P above $\frac{1}{2}$.

The principal wants early success to buy momentum, letting future rent differentials incentivize e_2 at a lower cost. She prioritizes winning early because doing so makes her more likely to win again, through the endogenous hot hand effect. Front-loading makes buying that early success cheaper, both directly, since more first-period budget lowers the cost of inducing e_1 , and indirectly,

²⁴Absent memory, terms (i) and (iii) alone determine b_1 , and they balance exactly at $b_1 = \frac{1}{2}$, matching the dynamic first-best spending level.

²⁵This ordering, $e_2(1) > e_2(0)$, holds for every interior e_1 , not only at the computed equilibrium (Appendix A.4).

since it widens the wedge that substitutes for the direct payment.²⁶ Budget and dynamic incentives work as complements, that is, budget makes the dynamic incentive tools work better.

3.6 Discussion

The introduction of a fixed budget produces three findings worth restating before unpacking their mechanics. First, hidden spending adds no distortion of its own in the static benchmark of Section 3.1. Principal and agent both choose $b = 1$, so hidden spending leaves only the standard hidden-effort wedge and no disagreement over the budget. Second, when the agent allocates the budget under hidden effort and hidden spending, he back-loads, that is, he spends less than half in the first period. Third, when the principal allocates the budget under hidden effort, she front-loads, spending more than half in the first period. The conflict over budget timing therefore is not simply because of asymmetric information, which is present in the static case as well, but rather due to continuation rents. The rent accrues to the agent, but the principal pays it, and under limited liability it is her cost of inducing effort. The same rent therefore enters their problems differently, and this asymmetry is what drives the opposite distortions. Table 1 documents the spending and effort levels behind these three findings. The payoffs and implementing transfers are provided in the propositions of Section 3.

Table 1: Equilibrium spending and effort across contracting environments at $n = \frac{1}{2}$. Details of all the numerical calculations can be found in Appendix B.

Case	b_1	e_1	$e_2(1)$	$e_2(0)$
Static first-best	1.00	0.71	–	–
Static second-best	1.00	0.35	–	–
Dynamic first-best	0.50	0.61	0.61	0.61
Agent controls spending	0.33	0.34	0.59	0.25
Principal controls spending	0.53	0.34	0.61	0.24

Asymmetry in the value of money. The principal and the agent face a similar intertemporal trade-off but resolve it in opposite directions. The trade-off is simple as spending more in the first period leaves less for the second. Both parties face the same technology, and memory holds in both

²⁶To contextualize this pattern, consider the timing of spending on new employees. Firms often spend more on a new hire early than later, on tools, licenses, and support that make the work easier without producing output on their own. While some of this spending is necessary for the new hire to start working, the model suggests that the remainder may reflect an incentive decision on the firm’s part.

cases, so the puzzle is why they resolve the same trade-off differently. The answer is that spending in the first period has a direct effect and an indirect effect, and the agent only considers the direct effect when choosing b_1 . The direct effect of increasing b_1 is that it lowers the cost of first-period effort. The indirect effect runs through the contract, since a change in b_1 changes second-period continuation rents, which changes the transfer needed to induce first-period effort.²⁷

To see which effects enter whose problem, compare the two $[b_1]$ -FOCs,

$$\begin{aligned} \text{Agent:} \quad & \underbrace{\frac{e_1^2}{(1+b_1)^{n+1}}}_{\text{cost saved in first period}} = \underbrace{\frac{e_1 e_2(1)^2 + (1-e_1)e_2(0)^2}{(2-b_1)^{n+1}}}_{\text{cost saving forgone in second period}}, \\ \text{Principal:} \quad & 2 \underbrace{\frac{e_1^2}{(1+b_1)^{n+1}}}_{\text{cost saved in first period}} + \underbrace{\frac{e_1(e_2(1)^2 - e_2(0)^2)}{(2-b_1)^{n+1}}}_{\text{rent wedge}} = 2 \underbrace{\frac{e_1 e_2(1)^2 + (1-e_1)e_2(0)^2}{(2-b_1)^{n+1}}}_{\text{cost saving forgone in second period}}. \end{aligned}$$

The agent's condition contains the direct effect alone. He takes transfers as fixed when optimizing over b_1 , so the indirect channel does not affect his problem. His condition weighs the first-period cost saving against the expected second-period cost saving. Memory makes the expected second-period cost saving the larger of the two, leading him to spend more in the second period. The principal's condition contains that same comparison at twice the magnitude, for reasons I return to below, with an additional term. This additional term is the rent wedge and reflects the indirect effect. She sets $t_1(1)$ knowing that it moves with b_1 , so she internalizes the full benefit of increasing first-period spending.

The rent wedge accounts for the entire gap between b_1^A and b_1^P .²⁸ Evaluated at the agent's optimum, where his condition holds with equality, the principal's derivative reduces to the wedge alone. Memory implies $e_2(1) > e_2(0)$, so the wedge is positive and the principal prefers more first-period spending than the agent. Memory drives both spending decisions, in the sense that it determines the effort profile. However, the asymmetry arises from which part of the effort profile each party cares about. The agent cares only about the expected level of squared effort. The principal cares about that as well, and also about the gap in effort levels between the branches. Effort levels, therefore, play a key role in determining how the parties resolve the trade-off.

²⁷This is the indirect channel of the principal's $[b_1]$ -FOC, terms (ii) and (iv) in Section 3.5.

²⁸Doubling both sides of a condition leaves its solution unchanged, so the shared comparison does not create the asymmetry.

Effort distortions across cases. As discussed above, memory holds in both the principal controls spending and the agent controls spending cases. Therefore, the standard dynamic moral-hazard pattern is obtained in both,

$$e_S^{FB} > e_{2D}^j(1) > e_{1D}^j > e_{2D}^j(0), \quad j \in \{P, A\}.$$

However, hidden spending creates a genuine additional effort distortion in the dynamic setting that is absent in the static setting. All three effort levels $(e_1, e_2(1), e_2(0))$ differ between the principal-controls and agent-controls cases even when b_1 is held fixed at the same value. The source is the agent's budget IC in (18), which binds only when the agent controls spending. This additional constraint involves all three transfers t_1 , $t_2(1, 1)$, and $t_2(0, 1)$, so satisfying it moves all three away from their principal-controls values. Each effort level is pinned down by its respective transfer through the standard effort IC, so all three induced effort levels shift accordingly.²⁹

The dynamic effort distortion thus has two distinct sources. First, the equilibrium b_1 differs across the two cases, and at each case's own optimum the implied effort levels naturally differ. Second, even conditional on the same b_1 , the agent's budget IC reshapes the optimal transfer profile and thereby the equilibrium effort levels. This contrasts with the static case where hidden spending adds no distortion beyond the standard moral-hazard wedge.

Cost of delegation. The principal's profit is strictly higher when she controls the spending than when the agent controls it.³⁰ Thus, centralization is optimal for the principal in the model. Delegation of budget control is nonetheless common in practice.³¹ The literature rationalizes delegation through frictions this model omits. The agent may hold private information about project costs or local conditions that would be expensive for the principal to elicit (Dessein, 2002). Attention or or-

²⁹Numerically, holding b_1 fixed at the principal-controls optimum b_1^P and re-solving the agent-controls problem with b_1 imposed via the agent's budget IC demonstrates that the implementing transfers shift, and the induced effort levels move with them. First-period effort rises from 0.34 to 0.37, $e_2(1)$ falls from 0.61 to 0.51, and $e_2(0)$ falls from 0.24 to 0.22. These values are computed in Appendix B.

³⁰The profit values are computed in Appendix B.

³¹For example, federal appropriations typically arrive as lump sums covering a set of related programs, and agencies themselves decide how to allocate and reallocate those funds across the programs and activities the appropriation covers (Congressional Research Service, 2024). Within the Department of Defense, budget authority passes from the Comptroller to the Military Departments and Defense Agencies, and from there to major commands and program executive offices, which sub-allocate to subordinate program offices for execution. The document releasing an allotment sets an amount that may not be exceeded (Defense Acquisition University, 2026).

ganizational constraints may prevent the principal from managing every budget allocation directly. Delegation may itself encourage the agent to gather information and propose projects, by committing the principal not to override him (Aghion and Tirole, 1997). This paper contributes a cost of delegation that those accounts do not contain, namely the back-loading distortion. The agent here holds no private information about the environment, so the cost is one of hidden action.³² Adding the omitted frictions would create a genuine trade-off, and whether the benefits of delegation still dominate once the back-loading cost is accounted for is left for future work. Delegation should dominate centralization when the agent’s private information is large relative to the back-loading cost. The cost grows as spending becomes more effective at reducing effort cost (Section 5.1), while the value of local information varies independently of it.

Tractability. The model features two hidden actions, spending and effort, in each period. Multi-dimensional hidden actions generally break implementability, and problems of this kind are rarely tractable. When two hidden choices both affect output, the principal cannot tell from observing output which action drove the result. Specifically, Holmstrom and Milgrom (1991) show that the agent can substitute between them in ways the principal cannot detect, and the incentive constraints become entangled. Therefore, the two actions cannot be solved for separately. Fernandes and Phelan (2000) show that dynamics makes this problem worse, since the additional action would then have to be tracked as a separate state variable.

This model sidesteps both problems and remains solvable through one feature. Spending does not affect output probabilities, it lowers only the cost of effort.³³ The effort IC therefore remains standard, and transfers pin down effort exactly as in a one-dimensional moral hazard problem, regardless of how much the agent has spent. Once effort is incentivized through the contract, the agent’s optimal budget choice follows as a separate best response.

The second concern is whether spending must be carried as an additional state variable alongside promised utility. In the second period, there is no future period left to save for. Since spending is costless, the agent exhausts whatever remains: $b_2 = 1 - b_1$ follows at every history. The fixed

³²The cost could be eliminated by making b_1 verifiable, meaning that the principal can condition payment on it, since she could then require her preferred allocation through a forcing contract. Delegation can still persist in practice because verification is costly or unavailable.

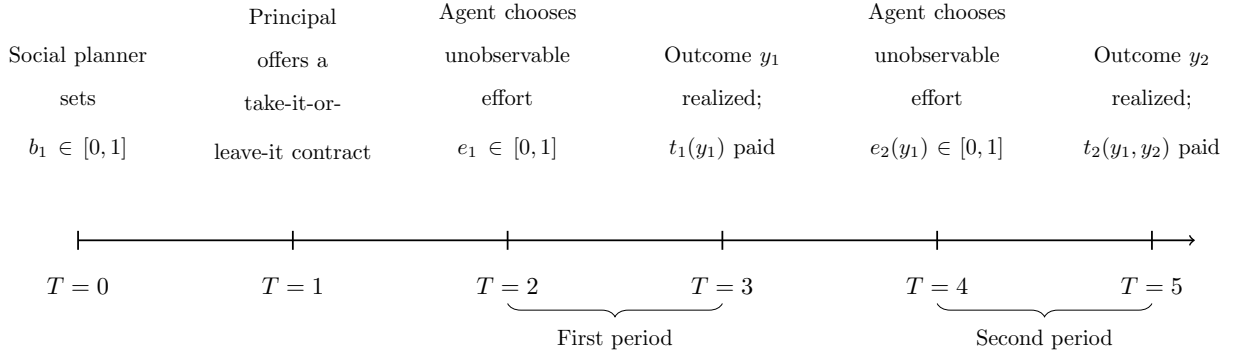
³³This assumption is also what isolates the mechanism behind the spending distortions. If money were productive, the effort IC would depend on spending, entangling the two constraints. Budget timing would also be distorted for reasons unrelated to incentives, since spending would raise success probabilities on its own.

budget, therefore, leaves only one spending decision. The agent chooses b_1 together with e_1 , before the first-period outcome is known. The agent's spending choice cannot depend on any history, and the same amount of budget remains whether the first period is a success or a failure. Spending does not differ across histories and needs no separate tracking. The state remains one-dimensional, as in the canonical setup of Spear and Srivastava (1987).

4 Social Planner and Welfare

Sections 3.4 and 3.5 establish that the principal and the agent disagree over how the budget should be split across periods. In this section, I consider a social planner who sets the budget allocation b_1 at the outset as a benchmark, rank the three allocations, and discuss policy implications.

4.1 Social planner benchmark



The social planner sets b_1 at $T = 0$, and then the standard dynamic moral hazard game between the principal and the agent unfolds. The figure above illustrates the timing of the new game. The planner's only instrument is the budget allocation. The continuation game has the principal designing the transfer schedule and the agent choosing effort, exactly the game of Section 3.5 but with b_1 imposed from outside. This is not in the agent controls spending regime as the agent has no spending decision to make. The planner maximizes total surplus

$$W(b_1) = \Pi(b_1) + U(b_1),$$

the sum of the principal's profit and the agent's rent at the resulting equilibrium.

Proposition 5 (Social planner). *At $n = \frac{1}{2}$ the social planner's optimal first-period spending is $b_1^{SP} \approx 0.56$. The resulting game between the principal and the agent yields*

$$e_1^{SP} \approx 0.35, \quad e_2^{SP}(1) \approx 0.60, \quad e_2^{SP}(0) \approx 0.24.$$

I solve the planner's problem numerically. The proof is in Appendix A.5 and the numerical procedure is in Appendix B. Combining Proposition 5 with Propositions 3 and 4 gives the ranking

$$b_1^A \approx 0.33 < b_1^P \approx 0.53 < b_1^{SP} \approx 0.56.$$

The social planner front-loads more than the principal, who front-loads more than the agent. The planner and the principal agree on which period should receive the larger share, but not on how large that share should be. The principal stops at b_1^P because her own profit is maximized there. A further increase in b_1 would still raise the agent's rent, but the principal ignores that gain, since it accrues to the agent rather than to her. The planner internalizes that rent increase, since it is part of total surplus and the planner does not care who captures it. Evaluating the planner's derivative at the principal's own optimum makes this precise: $d\Pi/db_1 = 0$ by definition, so dW/db_1 reduces to dU/db_1 , which is strictly positive. The planner pushes further to capture that marginal rent. In other words, the principal treats the agent's rent as a cost and the planner treats it as welfare.

The agent sits furthest from the planner. He weighs only his own effort cost, so he saves budget for the second period, where memory makes expected effort largest. The next section separates the forces producing this ordering.

4.2 Three channel framework

The three allocations can be organized by decomposing the marginal value of first-period spending into three channels. Each party sees a different subset, and at different weights, and the ranking follows from both.

The cost channel. This is the effort-cost effect of b_1 . Increasing b_1 lowers the cost of first-period effort and raises the cost of second-period effort. The net effect favors back-loading through

memory, which makes expected second-period effort larger than first-period effort, both in level and in volatility across branches.³⁴ Therefore, the budget is worth more in the second period. This comparison holds for anyone weighing total effort cost, so the channel points the same direction for all three parties. However, the channel carries a different weight in each party's problem.

The agent bears his own effort cost, so he sees the channel once, while the principal values it at twice the magnitude. The principal is forced to leave the agent rent due to hidden effort, and limited liability prevents her from taking it back. She pays for the agent's effort cost twice, once in compensating the cost and once as rent. Budget that lowers the effort cost therefore saves her twice, once as a cost saving and once as a rent saving. The planner puts the same weight on this channel as the agent, since the effort cost reduces surplus while the rent cancels as a transfer between the two parties.

The output channel. Increasing b_1 raises e_1 , and e_1 is the probability of reaching the good branch where second-period effort is highest. A higher b_1 buys a better chance of success in both periods through the endogenous hot hand effect, so this channel favors front-loading. The agent does not account for this channel, since output accrues to the principal.

The principal does value output, but the channel does not affect her b_1 decision. She already chooses efforts optimally, so at her optimum one more unit of effort is worth exactly what she pays to induce it. When b_1 shifts effort, the output she gains is offset by the transfers she must pay for it.

The channel survives only in the planner's b_1 choice. She inherits the efforts the principal chose, which are too low from a surplus standpoint, because the principal counts as cost a rent that cancels in welfare. The planner therefore has room to raise first-period effort further, which she does by shifting b_1 upward.

The wedge channel. As Section 3.6 shows, a wider second-period rent wedge substitutes for $t_1(1)$, letting the principal cut the direct transfer she pays for first-period success. Holding efforts fixed, raising b_1 widens the wedge, so this channel favors front-loading. Only the principal accounts for this channel when choosing b_1 , as the agent takes transfers as fixed and so does not consider

³⁴Section 3.4 discusses this. Since $e_2(1)$ is induced above $e_2(0)$, second-period effort is volatile across branches, which a convex cost function penalizes.

the saving on $t_1(1)$. The planner ignores it because the wedge is simply a redistribution between the two parties and cancels in $W = \Pi + U$.

To summarize, the cost channel favors back-loading, while the output and wedge channels favor front-loading. The cost and wedge channels are the effect of b_1 at fixed efforts. The output channel is the effect that runs through the efforts themselves. Therefore, it appears only in the planner's problem, since she both values the output and has not already internalized the effort responses. Table 2 shows which channels enter whose problem, and at what weight.

Table 2: The three channels in each party's choice of b_1 .

	Cost (back-loading)	Output (front-loading)	Wedge (front-loading)
Agent	$1\times$	–	–
Principal	$2\times$	nets to zero	$1\times$
Planner	$1\times$	$1\times$	–

The agent has only the cost channel, which pushes toward back-loading, so he back-loads. Both the principal and the planner have a front-loading channel pushing against it, and both therefore spend more than the agent in the first period.

The planner front-loads the most. She weighs the cost channel the same as the agent, and the output channel pushes against it. The principal weighs the cost channel at twice the magnitude, which dampens her front-loading below the planner's.³⁵

4.3 Welfare comparison

In this section, I discuss the welfare implications of the different contracts and spending allocations. Only the cost and output channels of the three discussed above survive in welfare. By revealed preference, the planner does strictly better than the principal-controls equilibrium, since b_1^P is available to her and she chooses otherwise. The agent-controls regime cannot be compared this way. In that setup, the agent chooses b_1 himself, so the principal faces the additional budget incentive constraint (IC_{b_1}), and the contract that results differs from the one the planner takes as given. The comparison is therefore an open one, and the outcome is surprising. At the baseline,

³⁵Comparing the agent and the planner isolates the output channel. They weigh the cost channel identically, and neither sees the wedge, so the entire gap between b_1^A and b_1^{SP} is attributable to the output channel that only the planner accounts for.

with $n = \frac{1}{2}$, the agent-controls regime delivers the highest total surplus of the three, ahead of the planner and the principal in that order.³⁶

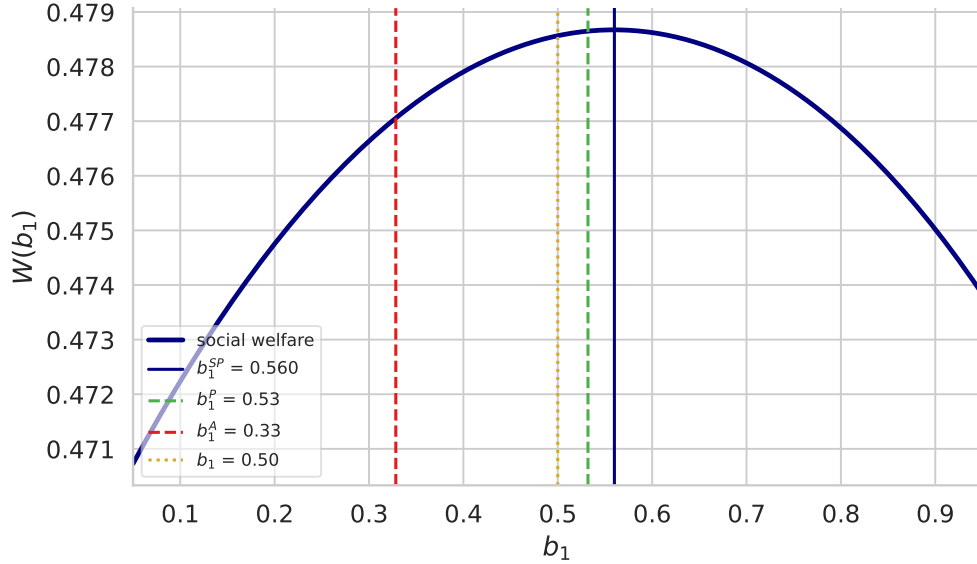


Figure 2: Welfare $W(b_1)$ under the principal-controls contract, with the three equilibrium allocations marked. The curve is single-peaked at $b_1^{SP} = 0.56$. The marker at $b_1^A = 0.33$ shows welfare under the principal's transfers at the agent's spending level, not W^A , which is attained under a different transfer profile and exceeds the peak of this curve.

The result needs some careful unfolding. On the instrument the planner controls, she does better than the agent by design, as Figure 2 confirms. The agent weighs only the cost channel while welfare weighs the output channel too, so his spending level leaves surplus on the table.

The difference between the two regimes lies elsewhere. The planner is constrained by the effort levels the principal's transfers induce, and first-period effort is too low from a surplus standpoint. As discussed previously, the principal distorts effort downwards from the social optimum to reduce rent paid to the agent. The planner counteracts this partly with the one instrument she controls, front-loading b_1 past the principal's choice. She cannot counteract it fully because she cannot set transfers, so the distortion in effort survives, and with it a loss of surplus.

The budget incentive constraint reaches the transfers themselves. A large $t_2(1, 1)$ raises $e_2(1)$ and expected second-period effort with it, which pushes the agent to back-load harder. The principal therefore wants to compress $t_2(1, 1)$, leaving a narrower wedge, which forces her to pay more directly for first-period success. As a result $t_1(1)$ rises, and at any given b_1 first-period effort rises with it,

³⁶The differences are small, and the ordering is specific to moderate n , as Section 5.1 demonstrates.

moving closer to the social optimum.

The surplus gained from that higher first-period effort outweighs both the lower failure-branch effort and the surplus lost to back-loading.³⁷ Therefore, a constraint that costs the principal profit raises total surplus. The planner cannot achieve the same, since she is restricted to a contract designed to maximize profit rather than surplus.

Section 3.6 presented the back-loading distortion as a new cost of delegation, to be weighed against the informational advantages that make delegation attractive. However, this section shows that the distortion is only a cost to the principal, not to society. In fact, for $n = \frac{1}{2}$, when money is moderately effective at reducing effort cost, it is welfare improving.

Interestingly, adding a second hidden action shrinks the effort distortion, while a new spending distortion appears. The second hidden action does not affect the principal's trade-off between inducing effort and limiting the rent she pays. However, it constrains her freedom to pick the three transfers independently. The profile she chooses to satisfy the budget incentive constraint pushes first-period effort up, reducing the downward distortion she would otherwise have chosen. In our model, the benefit from the smaller effort distortion dominates the cost of the spending distortion. This ordering changes as money becomes more effective at reducing effort cost, as Section 5.1 shows.

5 Extensions

5.1 Robustness in n : the value of money

Proposition 2 established that the dynamic first best splits the budget evenly for $n < 1$, and the results thus far use $n = \frac{1}{2}$. In this extension, I sweep n over that interior range and re-solve each contracting regime, following the same numerical procedure as at the baseline.³⁸ It is worth recalling that n measures how strongly budget reduces the cost of effort. A higher n , thus, raises equilibrium effort across both periods and branches, and both front-loading channels strengthen with it. The wedge widens as first-period effort rises.³⁹ The output channel compounds, since a

³⁷Conditional on b_1 , the principal sets $e_2(1)$ at its first-best level, so compressing this effort has no first-order welfare cost.

³⁸The numerical details are in Appendix B.

³⁹The good-branch effort does not depend on e_1 , but the return to bad-branch effort is scaled by the probability that failure occurs. A higher e_1 makes that branch less likely, so the principal induces less effort there. With $e_2(1)$ fixed and $e_2(0)$ falling, the wedge widens.

higher n raises both the probability of reaching the good branch and the effort induced once there.

Budget ordering. The ordering $b_1^A < \frac{1}{2} < b_1^P < b_1^{SP}$ holds throughout the range, as Figure 3 shows. All three allocations rise with n , but for different reasons and at different rates.

The agent's spending rises the slowest. He weighs only the cost channel, which puts back-loading pressure on him through memory. The agent wants to spend more in the period where he expects to exert more effort. As n rises, spending does more to reduce effort cost, so a smaller tilt toward the second period is enough to equalize the value of a unit of budget across the two periods. Therefore, the agent needs to back-load less to achieve the same result. This moves his spending toward the even split of the first best, but always remains below it, since memory leaves money more valuable in the second period at every n .

The principal and the planner were front-loading to begin with, so their rise moves them further from the first best. The principal's wedge channel strengthens, and the back-loading pressure weakens as n rises. However, she values that pressure at twice the magnitude, so it still pushes against her front-loading and her first-period spending rises only moderately. The planner increases her spending the fastest, since she faces no rent-doubling drag and fully captures the compounding output channel.

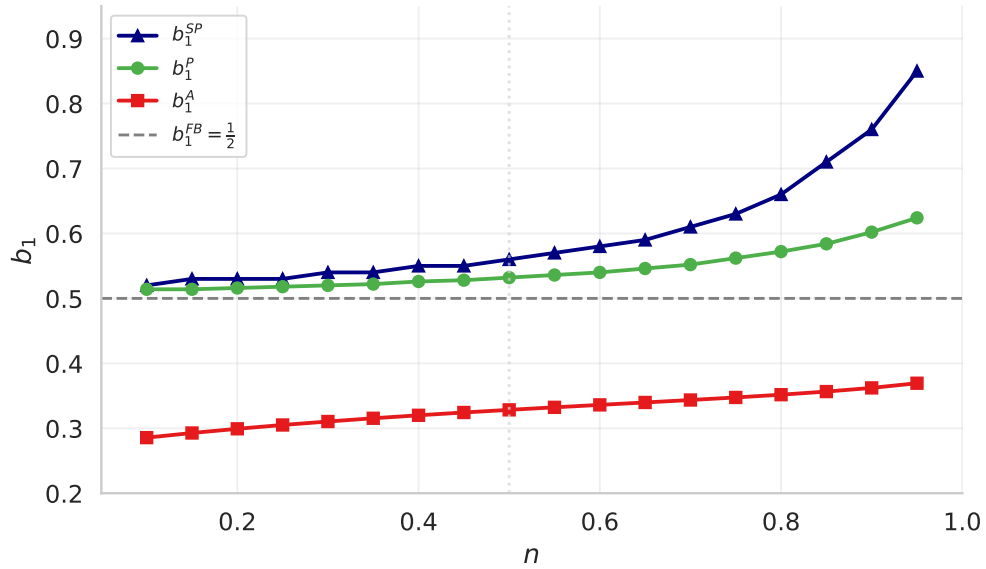


Figure 3: First-period spending across n in all three regimes. The ordering $b_1^A < \frac{1}{2} < b_1^P < b_1^{SP}$ is preserved throughout.

The principal's profit loss from delegation rises monotonically with n (Figure 4). Under delegation, she gets b_1^A rather than her preferred b_1^P , and that gap costs her more as n rises. Effort responds more strongly to budget at higher n , so the same spending gap displaces the induced effort levels much further from the principal's optimum. The case for centralization from the principal's point of view strengthens as money becomes more effective at reducing effort cost.

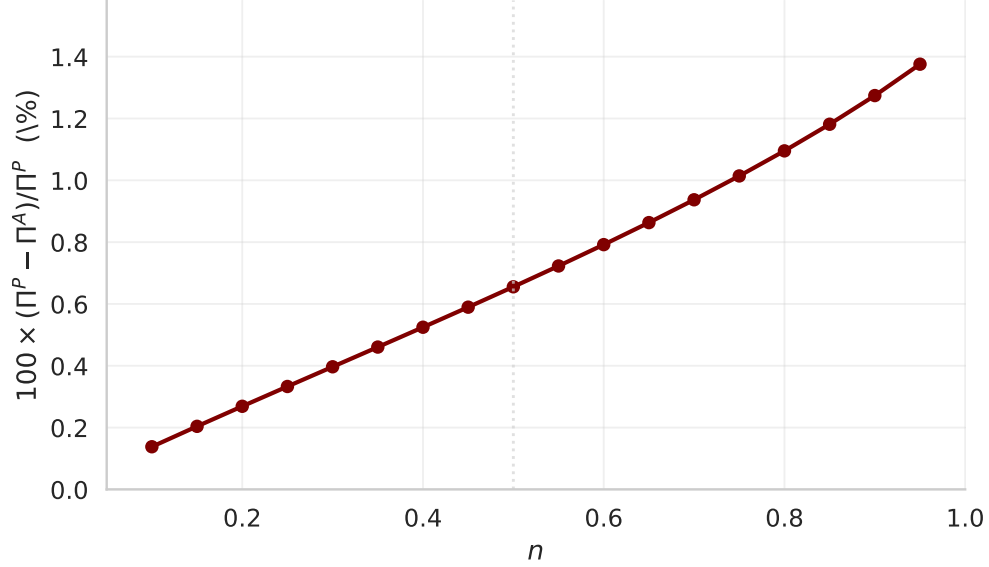


Figure 4: The principal's profit loss under delegation, as a percentage of her profit under principal controls.

Welfare ordering. At the baseline, the welfare ordering is $W^A > W^{SP} > W^P$. Across the sweep, $W^{SP} \geq W^P$ is preserved by revealed preference, but the position of W^A shifts, as Figure 5 traces.

Section 4.3 showed that welfare is highest when the agent controls spending, because at any given budget allocation, his budget incentive constraint pushes first-period effort toward the level that maximizes surplus. The gain from the reduced effort distortion outweighs the surplus forgone by his low spending. As n rises, the surplus forgone grows sharply. The output channel compounds, since a higher n raises both the probability of reaching the good branch and the value of being there. Therefore, the surplus return to front-loading grows with it. The agent does not care about the output channel at all, so his spending stays low while the planner's rises sharply. The surplus he forgoes by back-loading eventually overtakes the gain from the reduced effort distortion, and the

welfare ordering flips. The planner overtakes the agent first, since she captures the output channel directly and front-loads aggressively to do so. The principal follows later, as she front-loads only moderately compared to the planner.

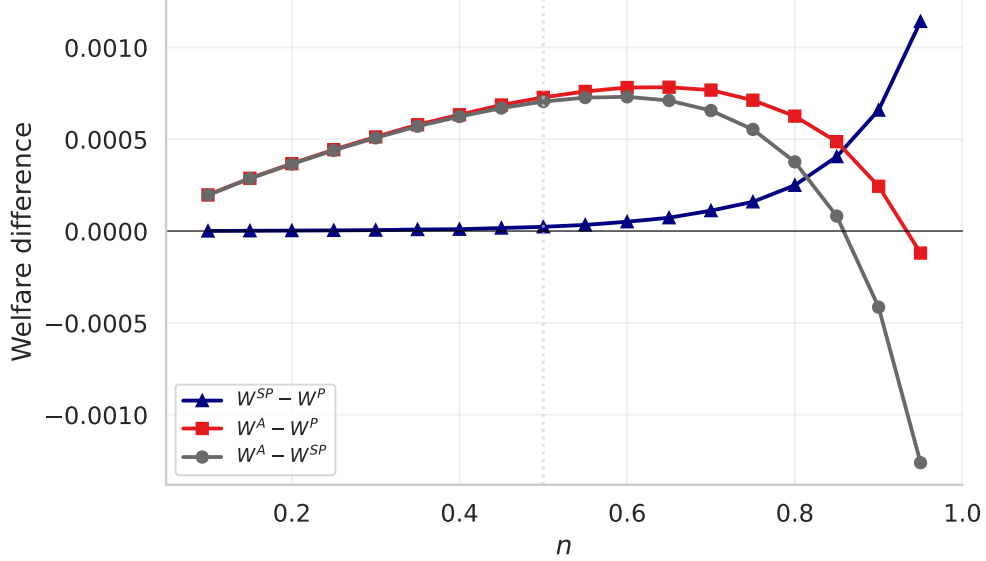


Figure 5: Welfare differences across n . The agent controls spending advantage disappears as spending becomes more effective at reducing effort cost.

To summarize, the welfare advantage when the agent controls spending is not a knife-edge result, as it survives over a substantial range around the baseline. However, as money becomes more effective at reducing effort cost, the case for delegation weakens without considering any of its informational advantages.

5.2 Discounting

Adding discounting to the baseline model does not produce novel results, and I include it simply for completeness. As intuition suggests, a lower discount factor makes second-period payoffs worth less, so both parties shift spending toward the first period. Placing $\delta \in (0, 1]$ on second-period payoffs leaves the qualitative results unchanged, and the ordering $b_1^A < b_1^P < b_1^{SP}$ is preserved. As δ falls, both the principal controls and agent controls spending regimes collapse to the entire budget being spent in the first period.

The one result worth noting is how quickly the principal moves to spending the entire budget in the first period as the future becomes less valuable. The principal reaches $b_1^P = 1$ at a discount

factor well above zero, as Figure 6 shows. She front-loads already at $\delta = 1$, so a modest reduction in the weight on second-period payoffs is enough to eliminate the back-loading pressure that keeps her allocation interior. I solve the principal controls spending case explicitly and establish the limiting result for the agent controls spending case; Appendix A.6 provides the details.

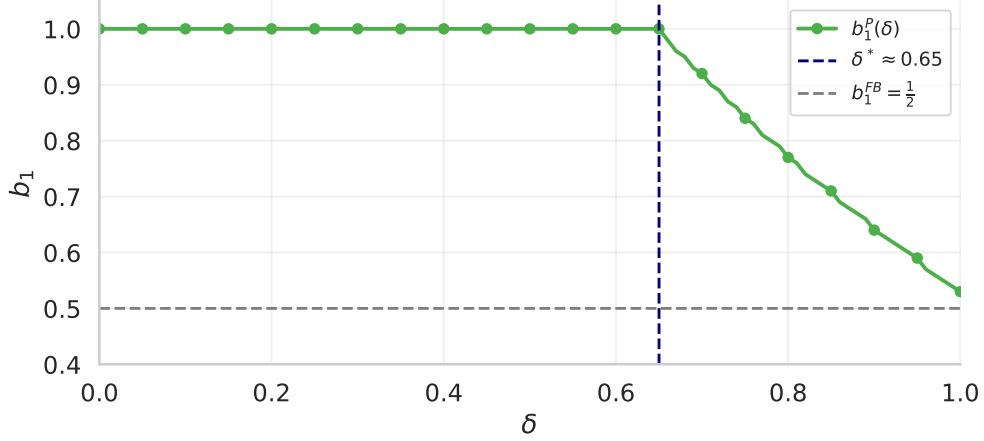


Figure 6: First-period spending under principal controls spending across the discount factor. The allocation sits at the corner for δ below $\delta^* \approx 0.65$ and becomes interior above it.

6 Conclusion

This paper studies a tractable model of hidden effort and intertemporal budget allocation, and shows that dynamic incentives alone can distort the timing of spending. In particular, the paper compares two contracting environments, one where the principal allocates the fixed budget and one where the agent does. Memory holds in both, as in the standard dynamic problem: second-period effort depends on the first-period outcome, and is induced higher after a success. This property of the optimal contract distorts intertemporal spending away from the first-best, and the direction depends on who controls the spending. The principal front-loads, spending more than half the budget in the first period, since this widens the second-period rent wedge and lets her reduce the direct transfer she needs to pay for early success. The agent back-loads instead, since budget is worth more in the second period where effort is highest.

As a byproduct of this divergence, delegating the budget costs the principal profit. This is a new cost, and it should be weighed against the informational advantages that usually justify delegation.

Additionally, under the period-as-phase interpretation of Remark 1, the agent’s back-loading offers a novel mechanism for budget hoarding and the late-period spending surges documented empirically. The spike here arises from memory in the optimal contract alone, without the budget rules or agency frictions that the existing literature relies on.

Introducing a social planner who fixes the budget allocation, I find that the planner front-loads more than the principal. The planner’s allocation, together with the principal’s and the agent’s, can be explained through three channels. A cost channel favors back-loading, while wedge and output channels favor front-loading. The agent weighs only the cost channel, so he back-loads. The principal and the planner each have a front-loading channel that outweighs the cost channel. The principal falls short of the planner because she weighs the cost channel twice, once as compensation and once as rent.

The paper has a surprising welfare consequence. When money is moderately effective at reducing effort cost, delegation raises total surplus even though it lowers the principal’s profit. The gain results from satisfying the agent’s budget incentive constraint, which forces the principal to choose transfers that leave first-period effort less distorted, rather than from the agent’s spending level itself. The cost of delegation identified above is, therefore, a cost to the principal rather than to society. However, this welfare advantage vanishes as money becomes more effective at reducing effort cost.

The paper has several limitations that point to natural extensions. First, I consider a two-period model for tractability. With even three periods, the agent makes a second allocation decision after observing the first-period outcome, so the budget entering the final period can differ across branches and would need to be tracked. Solving the general T -period model is left as future work, but I expect the recursive logic to hold with the larger state and both distortions to compound as the horizon grows.

The model also assumes that the agent’s spending is not contractible. If the principal could condition payment on it, she could enforce her preferred allocation through a forcing contract. Verification is rarely free, however, and I expect the equilibrium allocation to fall between b_1^A and b_1^P depending on the cost of verification. The optimal contract under costly verification is left for future work.

I also assume that the principal has full commitment. If renegotiation cannot be prevented, the

bad-branch stick disappears while the good-branch carrot survives. This narrows the rent wedge and makes the second period more attractive to both principal and agent. Therefore, I conjecture that the principal's front-loading distortion will shrink and the agent's back-loading distortion will grow, as both shift spending toward the second period. However, in this model, the renegotiation-proof floor on the agent's continuation payoff is itself a function of the budget allocation. This adds terms to both the principal's and the agent's first-order conditions making the characterization more complex, so a full treatment is beyond the scope of this paper.

Finally, the model abstracts from private information about productivity or project quality, which is the focus of much of the delegation literature. A unified treatment could ask whether delegation's informational benefits are large enough to outweigh the profit cost identified above.

A Proofs

A.1 Proof of Proposition 1

This appendix gives the full derivation for all five static cases.

First-best (§3.1.1). The principal solves

$$\max_{e, b \in [0,1]} e - \frac{e^2}{(1+b)^n}.$$

The effort first-order condition is

$$[e] : \quad 1 - \frac{2e}{(1+b)^n} = 0 \quad \implies \quad e^*(b) = \frac{(1+b)^n}{2}.$$

Substituting back,

$$\Pi(b) = \frac{(1+b)^n}{4},$$

which is strictly increasing in b for $n > 0$, so $b_S^{\text{FB}} = 1$. Then

$$e_S^{\text{FB}} = \frac{2^n}{2}, \quad \Pi_S^{\text{FB}} = \frac{2^n}{4}.$$

The IR constraint binds, so the transfer equals the agent's effort cost,

$$t_S^{\text{FB}} = \frac{(e_S^{\text{FB}})^2}{2^n} = \frac{2^n}{4}, \quad A_S^{\text{FB}} = 0. \quad \square$$

Principal controls spending, effort unobservable (§3.1.2). Limited liability binds $t(0) = 0$.

The agent's effort IC is

$$[e] : \quad t(1) - \frac{2e}{(1+b)^n} = 0 \quad \implies \quad t(1) = \frac{2e}{(1+b)^n}.$$

Substituting into profit $\Pi = e(1 - t(1))$, the principal solves

$$\max_{e, b \in [0,1]} e - \frac{2e^2}{(1+b)^n},$$

with effort first-order condition

$$[e] : \quad 1 - \frac{4e}{(1+b)^n} = 0 \quad \implies \quad e^*(b) = \frac{(1+b)^n}{4}.$$

Substituting back,

$$\Pi(b) = \frac{(1+b)^n}{8},$$

strictly increasing in b , so $b_S^P = 1$. Then

$$e_S^P = \frac{2^n}{4}, \quad t_S^P = \frac{1}{2}, \quad \Pi_S^P = \frac{2^n}{8}.$$

The agent's rent is

$$A_S^P = e t(1) - \frac{e^2}{(1+b)^n} = \frac{2^n}{8} - \frac{2^n}{16} = \frac{2^n}{16}. \quad \square$$

Agent controls spending, both unobservable (§3.1.3). The agent jointly chooses (e, b) ,

$$\max_{e, b} e t(1) - \frac{e^2}{(1+b)^n},$$

with effort first-order condition

$$[e] : \quad t(1) - \frac{2e}{(1+b)^n} = 0 \quad \implies \quad e = \frac{t(1)(1+b)^n}{2}.$$

Substituting back, the agent's payoff is

$$U(b) = e t(1) - \frac{e^2}{(1+b)^n} = \frac{t(1)^2(1+b)^n}{4},$$

strictly increasing in b , so $b = 1$. Anticipating this, the principal solves

$$\max_e e(1 - t(1)) = \frac{t(1)(1+b)^n}{2} (1 - t(1)) \Big|_{b=1},$$

with first-order condition in the transfer

$$[t] : \quad \frac{2^n}{2} (1 - 2t(1)) = 0 \quad \implies \quad t_S^{\text{SB}} = \frac{1}{2}.$$

This is the standard hidden-effort problem of §3.1.2, giving

$$e_S^{\text{SB}} = \frac{2^n}{4}, \quad \Pi_S^{\text{SB}} = \frac{2^n}{8}, \quad A_S^{\text{SB}} = \frac{2^n}{16}. \quad \square$$

Agent controls, effort observable / spending unobservable. Facing a schedule $t(e)$ contingent on observed effort, the agent chooses effort and spending jointly,

$$\max_{e, b \in [0,1]} t(e) - \frac{e^2}{(1+b)^n}.$$

For any effort, the cost is strictly decreasing in b , so the agent sets $\tilde{b} = 1$. The principal anticipates this and induces her preferred effort at its cost, with the IR binding at $t(e) = \frac{e^2}{2^n}$, so her problem is

$$\max_e e - \frac{e^2}{2^n} \quad \implies \quad e = \frac{2^n}{2}.$$

This reproduces the first best of §3.1.1. □

Agent controls, spending observable / effort unobservable. For any b and posted $t(b)$, the agent's effort IC is

$$[e] : \quad t(b) - \frac{2e}{(1+b)^n} = 0 \quad \implies \quad e(b) = \frac{t(b)(1+b)^n}{2},$$

giving payoff

$$U(b) = \frac{t(b)^2(1+b)^n}{4}.$$

The agent chooses b to maximize $U(b)$ given the posted schedule, and the principal posts $t(b)$ anticipating that choice. The principal maximizes

$$\Pi = e(1 - t(b)) = \frac{t(b)(1+b)^n}{2}(1 - t(b)),$$

with pointwise first-order condition in the transfer

$$\frac{(1+b)^n}{2}(1 - 2t(b)) = 0 \quad \implies \quad t^*(b) = \frac{1}{2},$$

independent of b . Then $e(b) = \frac{(1+b)^n}{4}$ and $\Pi(b) = \frac{(1+b)^n}{8}$, and $U(b) = \frac{(1+b)^n}{16}$ is strictly increasing, so the agent chooses $b = 1$. The induced outcome is the second best,

$$t_S^{\text{SB}} = \frac{1}{2}, \quad e_S^{\text{SB}} = \frac{2^n}{4}, \quad \Pi_S^{\text{SB}} = \frac{2^n}{8}, \quad A_S^{\text{SB}} = \frac{2^n}{16},$$

identical to §3.1.2. □

A.2 Proof of Proposition 2 (Dynamic first-best)

With effort and spending observable, IR binds and the principal maximizes

$$e_1 \left(1 + e_2(1) - \frac{e_2(1)^2}{(1+b_2(1))^n} \right) + (1 - e_1) \left(e_2(0) - \frac{e_2(0)^2}{(1+b_2(0))^n} \right) - \frac{e_1^2}{(1+b_1)^n}$$

$$\text{subject to} \quad b_2(y_1) \leq 1 - b_1.$$

The budget constraint binds since $c(e, b)$ is strictly decreasing in b for $e > 0$, so $b_2(y_1) = 1 - b_1$.

The objective becomes

$$e_1 \left(1 + e_2(1) - \frac{e_2(1)^2}{(2 - b_1)^n} \right) + (1 - e_1) \left(e_2(0) - \frac{e_2(0)^2}{(2 - b_1)^n} \right) - \frac{e_1^2}{(1 + b_1)^n}.$$

The second-period FOCs are

$$[e_2(y_1)] : \quad 1 - \frac{2e_2(y_1)}{(2 - b_1)^n} = 0 \quad \implies \quad e_2^{\text{FB}}(y_1) = \frac{(2 - b_1)^n}{2}, \quad y_1 \in \{0, 1\}.$$

Substituting these in, the first-period FOC is

$$[e_1] : \quad 1 - \frac{2e_1}{(1 + b_1)^n} = 0 \quad \implies \quad e_1^{\text{FB}} = \frac{(1 + b_1)^n}{2}.$$

Plugging this back into the objective, the reduced form in b_1 alone is

$$\Pi(b_1) = \frac{(1 + b_1)^n}{2} - \frac{(1 + b_1)^n}{4} + \frac{(2 - b_1)^n}{4} = \frac{(1 + b_1)^n}{4} + \frac{(2 - b_1)^n}{4},$$

with first and second derivatives given by:

$$\Pi'(b_1) = \frac{n}{4} [(1 + b_1)^{n-1} - (2 - b_1)^{n-1}], \quad \Pi''(b_1) = \frac{n(n-1)}{4} [(1 + b_1)^{n-2} + (2 - b_1)^{n-2}].$$

The curvature of Π determines the optimal b_1 . The bracketed terms in Π'' are positive, so the sign of Π'' is that of $n - 1$. For $n < 1$, Π is strictly concave and $\Pi'(\frac{1}{2}) = 0$ by symmetry, so the maximum is interior at $b_{1D}^{\text{FB}} = \frac{1}{2}$. For $n > 1$, Π is strictly convex, yielding a maximum at a corner, $b_{1D}^{\text{FB}} \in \{0, 1\}$. For $n = 1$, $\Pi \equiv \frac{3}{4}$ on $[0, 1]$, so any allocation is optimal.

Fixing $n = \frac{1}{2}$, the concavity above gives $b_{1D}^{\text{FB}} = \frac{1}{2}$, and plugging this into (8) yields the baseline outcome

$$b_{1D}^{\text{FB}} = \frac{1}{2}, \quad e_{1D}^{\text{FB}} = e_{2D}^{\text{FB}}(1) = e_{2D}^{\text{FB}}(0) = \frac{\sqrt{6}}{4} \approx 0.612, \quad \Pi_{1D}^{\text{FB}} = \frac{\sqrt{6}}{4} \approx 0.612,$$

with $A_{1D}^{\text{FB}} = 0$ since IR binds and the agent earns no rent under full information. $\square$

A.3 Proof of Proposition 3 (Agent controls spending contract)

The agent's value at a given b_1 , optimizing over e_1 , is $V(b_1) = U(e_1^*(b_1), b_1)$ where

$$U(e_1, b_1) = e_1 a(1) + (1 - e_1) a(0) - \frac{e_1^2}{(1 + b_1)^n}.$$

By the envelope theorem,

$$\frac{dV}{db_1} = \frac{n e_1^2}{(1 + b_1)^{n+1}} + e_1 \frac{\partial a(1)}{\partial b_1} + (1 - e_1) \frac{\partial a(0)}{\partial b_1}.$$

Using (10) and $b_2(y_1) = 1 - b_1$,

$$\frac{\partial a(y_1)}{\partial b_1} = -\frac{n e_2(y_1)^2}{(2 - b_1)^{n+1}},$$

so

$$\frac{dV}{db_1} = \frac{n e_1^2}{(1 + b_1)^{n+1}} - \frac{n [e_1 e_2(1)^2 + (1 - e_1) e_2(0)^2]}{(2 - b_1)^{n+1}},$$

which is (IC_{b_1}) . With the failure transfers at zero, $(IC_2(y_1))$ reduces to

$$e_2(y_1) = \frac{(2 - b_1)^n}{2} t_2(y_1, 1),$$

and e_1 from (12) is

$$e_1 = \frac{(1 + b_1)^n}{2} t_1(1) + \frac{(1 + b_1)^n (2 - b_1)^n}{8} [t_2(1, 1)^2 - t_2(0, 1)^2].$$

Substituting both into $\frac{dV}{db_1} = 0$ and clearing denominators gives

$$\begin{aligned}
0 = & (1 + b_1)^{n-1} t_1(1)^2 \\
& + \frac{(1 + b_1)^{n-1} (2 - b_1)^n}{2} \left(t_1(1) t_2(1, 1)^2 - t_1(1) t_2(0, 1)^2 \right) \\
& + \frac{(1 + b_1)^{n-1} (2 - b_1)^{2n}}{8} \left(\frac{1}{2} t_2(1, 1)^4 + \frac{1}{2} t_2(0, 1)^4 - t_2(1, 1)^2 t_2(0, 1)^2 \right) \\
& - \frac{(1 + b_1)^n (2 - b_1)^{n-1}}{2} \left(t_1(1) t_2(1, 1)^2 - t_1(1) t_2(0, 1)^2 \right) \\
& + \frac{(1 + b_1)^n (2 - b_1)^{2n-1}}{4} \left(t_2(1, 1)^2 t_2(0, 1)^2 - \frac{1}{2} t_2(1, 1)^4 - \frac{1}{2} t_2(0, 1)^4 \right) \\
& - (2 - b_1)^{n-1} t_2(0, 1)^2.
\end{aligned} \tag{18}$$

Setting $t_1(1) = t_2(1, 1) = t_2(0, 1) = t$ collapses (18) to

$$t^2 [(1 + b_1)^{n-1} - (2 - b_1)^{n-1}] = 0,$$

confirming that $b_1 = \frac{1}{2}$ in the case when the contract displays no memory.

Substituting $e_2(y_1)$ and e_1 above into

$$\Pi = e_1 \left(1 + e_2(1) (1 - t_2(1, 1)) - t_1(1) \right) + (1 - e_1) e_2(0) (1 - t_2(0, 1))$$

and taking the $[t_1(1)]$ -FOC at fixed b_1 gives

$$t_1(1) = \frac{1}{2} + \frac{(2 - b_1)^n}{4} \left(t_2(1, 1) - t_2(0, 1) \right) + \frac{3(2 - b_1)^n}{8} \left(t_2(0, 1)^2 - t_2(1, 1)^2 \right).$$

The $[t_2(1, 1)]$ - and $[t_2(0, 1)]$ -FOCs are similarly nonlinear. Together with (18), this is a system of four equations in four unknowns with no closed-form solution. I solve it numerically: for each candidate transfer triple, b_1 is pinned down by (18), and profit is maximized over the triple by grid search refined locally, with uniqueness checked from multiple starting points. This yields

$$b_1^A \approx 0.33, \quad t_1(1) \approx 0.36, \quad t_2(1, 1) \approx 0.92, \quad t_2(0, 1) \approx 0.39,$$

$$e_{1D}^A \approx 0.34, \quad e_{2D}^A(1) \approx 0.59, \quad e_{2D}^A(0) \approx 0.25, \quad \Pi_{1D}^A \approx 0.333, \quad A_{1D}^A \approx 0.146.$$

□

A.4 Proof of Proposition 4 (Principal controls spending contract)

Substituting $t_2(y_1, 1)$ and $t_1(1)$ from the body into profit and simplifying gives (13). Differentiating with respect to $e_2(1)$, e_1 , $e_2(0)$, and b_1 gives the four FOCs stated in the body, including (14) and (15).

The $[e_2(0)]$ -FOC is linear in $e_2(0)$ and solves explicitly,

$$e_2(0) = \frac{(2 - b_1)^n}{2} \cdot \frac{1 - e_1}{2 - e_1}. \quad (19)$$

Comparing to (14),

$$\frac{e_2(0)}{e_2(1)} = \frac{1 - e_1}{2 - e_1} < 1 \quad \text{for every } e_1 \in (0, 1),$$

so memory, $e_2(1) > e_2(0)$, holds at every interior e_1 , not merely at the computed optimum. Substituting (19) into the $[e_1]$ -FOC leaves a single equation in e_1 and b_1 ,

$$1 + \frac{(2 - b_1)^n}{4} - \frac{4e_1}{(1 + b_1)^n} - \frac{(2 - b_1)^n}{2} \cdot \frac{1 - e_1}{2 - e_1} + \frac{(2 - b_1)^n}{4} \left(\frac{1 - e_1}{2 - e_1} \right)^2 = 0, \quad (20)$$

nonlinear in e_1 . Together with (15), this leaves a system of two equations in (e_1, b_1) with no closed-form solution.

I solve the system by sweeping $b_1 \in [0, 1]$ on a grid, solving for e_1 at each b_1 with $e_2(0)$ and $e_2(1)$ recovered from (19) and (14), and picking the b_1 that maximizes profit. This yields

$$b_1^P \approx 0.53, \quad e_{1D}^P \approx 0.34, \quad e_{2D}^P(1) \approx 0.61, \quad e_{2D}^P(0) \approx 0.24, \quad \Pi_{1D}^P \approx 0.336, \quad A_{1D}^P \approx 0.143,$$

with implementing transfers $t_1(1) \approx 0.30$, $t_2(1, 1) \approx 1.00$, and $t_2(0, 1) \approx 0.40$. □

A.5 Proof of Proposition 5 (Social planner)

Given a fixed b_1 , the principal's problem is unchanged from §3.5. She solves

$$\max_{t_1(1), t_2(1,1), t_2(0,1)} e_1(1 - t_1(1)) + e_1 e_2(1)(1 - t_2(1, 1)) + (1 - e_1) e_2(0)(1 - t_2(0, 1))$$

subject to limited liability, (12), and (IC₂(y₁)), yielding effort levels $e_1^*(b_1)$, $e_2^*(1; b_1)$, and $e_2^*(0; b_1)$ as functions of b_1 . Transfers cancel between principal and agent, so the planner's objective is total output net of total effort cost,

$$W(b_1) = e_1^* + e_1^* e_2^*(1) + (1 - e_1^*) e_2^*(0) - \frac{e_1^{*2}}{(1 + b_1)^n} - \frac{e_1^* e_2^*(1)^2 + (1 - e_1^*) e_2^*(0)^2}{(2 - b_1)^n}.$$

The ranking follows immediately. Writing $W = \Pi + U$ and evaluating at $b_1 = b_1^P$, the first term vanishes by the principal's optimum, so

$$\left. \frac{dW}{db_1} \right|_{b_1^P} = \left. \frac{d\Pi}{db_1} \right|_{b_1^P} + \left. \frac{dU}{db_1} \right|_{b_1^P} = \left. \frac{dU}{db_1} \right|_{b_1^P} > 0.$$

Welfare is still increasing at the principal's choice, so the planner's optimum lies strictly above b_1^P .⁴⁰ With $b_1^A < b_1^P$ from Propositions 3 and 4, this gives

$$b_1^A < b_1^P < b_1^{SP}.$$

The level of b_1^{SP} requires the planner's first-order condition. Since the efforts depend on b_1 , the total derivative is

$$\frac{dW}{db_1} = \underbrace{\left. \frac{\partial W}{\partial b_1} \right|_e}_{\text{direct}} + \underbrace{\sum_{j \in \{1, 2(1), 2(0)\}} \frac{\partial W}{\partial e_j^*} \frac{de_j^*}{db_1}}_{\text{indirect}}.$$

The direct effect holds efforts fixed, so only the two cost terms contribute,

$$\left. \frac{\partial W}{\partial b_1} \right|_e = \frac{n e_1^{*2}}{(1 + b_1)^{n+1}} - \frac{n [e_1^* e_2^*(1)^2 + (1 - e_1^*) e_2^*(0)^2]}{(2 - b_1)^{n+1}},$$

the saving on first-period effort cost against the forgone saving on second-period effort cost. This coincides in form with (IC_{b₁}), though here it is not an incentive constraint. The agent does not choose b_1 in this regime, so the expression enters only as the direct effect of spending on costs.

The indirect terms do not vanish, since the efforts are chosen optimally for the principal's

⁴⁰The sign is verified numerically at b_1^P and is local. The agent's rent is not monotone in b_1 over the whole range.

objective rather than the planner's. Substituting the principal's $[t_1(1)]$ -FOC,

$$(1 - t_1^*(1)) + e_2^*(1)(1 - t_2^*(1, 1)) - e_2^*(0)(1 - t_2^*(0, 1)) = \frac{2e_1^*}{(1 + b_1)^n},$$

into $\frac{\partial W}{\partial e_1}$ gives

$$[e_1] : \quad \frac{\partial W}{\partial e_1^*} = \frac{2e_1^*}{(1 + b_1)^n},$$

which is strictly positive and equals the agent's marginal cost of first-period effort. Substituting $(IC_2(y_1))$ into the second-period derivatives gives

$$[e_2(0)] : \quad \frac{\partial W}{\partial e_2^*(0)} = (1 - e_1^*) \left(1 - \frac{2e_2^*(0)}{(2 - b_1)^n} \right), \quad [e_2(1)] : \quad \frac{\partial W}{\partial e_2^*(1)} = e_1^*(1 - t_2^*(1, 1)).$$

The $[e_2(1)]$ -FOC of the principal's problem gives $e_2^*(1) = \frac{(2-b_1)^n}{2}$. Substituting this into $(IC_2(y_1))$,

$$t_2^*(1, 1) = \frac{2e_2^*(1)}{(2 - b_1)^n} = \frac{2}{(2 - b_1)^n} \cdot \frac{(2 - b_1)^n}{2} = 1$$

for every b_1 , so $\frac{\partial W}{\partial e_2^*(1)} = 0$ and the $e_2(1)$ channel drops out. The planner's first-order condition is therefore

$$\begin{aligned} \frac{dW}{db_1} &= \frac{n e_1^{*2}}{(1 + b_1)^{n+1}} - \frac{n[e_1^* e_2^*(1)^2 + (1 - e_1^*) e_2^*(0)^2]}{(2 - b_1)^{n+1}} \\ &\quad + \frac{2e_1^*}{(1 + b_1)^n} \frac{de_1^*}{db_1} + (1 - e_1^*) \left(1 - \frac{2e_2^*(0)}{(2 - b_1)^n} \right) \frac{de_2^*(0)}{db_1} = 0, \end{aligned} \quad (21)$$

with the first line the direct effect and the second the indirect effect.

Writing $\alpha = (1 + b_1)^n$ and $\beta = (2 - b_1)^n$, substituting (19) into the $[e_1]$ -FOC of the principal's problem and simplifying reduces (20) to

$$\frac{4e_1^*}{\alpha} = 1 + \frac{\beta}{4(2 - e_1^*)^2}, \quad (22)$$

and clearing denominators gives

$$-16e_1^{*3} + (4\alpha + 64)e_1^{*2} - (16\alpha + 64)e_1^* + \alpha(\beta + 16) = 0,$$

a cubic in e_1^* , whose roots can be found in closed form. Both $e_2^*(1)$ and $e_2^*(0)$ are given explicitly by (14) and (19), differentiating (22) implicitly gives $\frac{de_1^*}{db_1}$ in terms of e_1^* and b_1 , and differentiating (19) then gives $\frac{de_2^*(0)}{db_1}$. Every term in (21) is therefore explicit once e_1^* is substituted from the cubic, but α and β are fractional powers of different linear functions of b_1 , and the planner's optimal b_1 does not appear to admit a closed form.

I solve (21) numerically on a grid over $b_1 \in [0.05, 0.95]$. At each grid point I maximize the principal's profit over the transfer triple from several starting points, recover the induced efforts from the effort ICs, and evaluate $W(b_1)$. The planner's choice is the grid point with the highest welfare. This yields

$$b_1^{SP} \approx 0.56, \quad e_1^{SP} \approx 0.35, \quad e_2^{SP}(1) \approx 0.60, \quad e_2^{SP}(0) \approx 0.24, \quad W^{SP} \approx 0.479,$$

with implementing transfers $t_1(1) \approx 0.30$, $t_2(1, 1) \approx 1.00$, and $t_2(0, 1) \approx 0.40$. □

A.6 Discounting

With discount factor $\delta \in (0, 1]$ on second-period payoffs, the principal's first-period IC becomes

$$\frac{2e_1}{(1+b_1)^n} = t_1(1) + \delta \frac{e_2(1)^2 - e_2(0)^2}{(2-b_1)^n},$$

and every second-period term in the FOC system carries a factor δ . The $[e_2(1)]$ -FOC is unchanged, $e_2(1) = \frac{(2-b_1)^n}{2}$, since the δ cancels. The $[b_1]$ condition is

$$\frac{2e_1^2}{(1+b_1)^{n+1}} + \delta \frac{e_1 e_2(1)^2}{(2-b_1)^{n+1}} = \delta \frac{2e_1 e_2(1)^2 + 2e_2(0)^2 - e_1 e_2(0)^2}{(2-b_1)^{n+1}}.$$

As $\delta \rightarrow 0$ the right side vanishes while the left remains strictly positive, so no interior solution exists and $b_1^P = 1$. As $\delta \rightarrow 1$ the system reduces to that of Proposition 4, with $b_1^P \approx 0.53$. The difference between the two sides is continuous in δ , so there is a threshold δ^* at which the solution becomes interior. I locate it by repeating the sweep of Proposition 4 at each δ on a grid, solving the effort system at every b_1 and taking the profit maximizer. The threshold is the smallest δ at which that maximizer falls below one, which gives $\delta^* \approx 0.65$ at $n = \frac{1}{2}$.

For agent-controls, $(\mathbf{IC}_{b_1})$ becomes

$$\frac{dV}{db_1} = \frac{n e_1^2}{(1 + b_1)^{n+1}} - \delta \frac{n [e_1 e_2(1)^2 + (1 - e_1) e_2(0)^2]}{(2 - b_1)^{n+1}} = 0,$$

where the first-period cost saving is undiscounted and the forgone second-period saving carries δ . As $\delta \rightarrow 0$ the second term vanishes, the derivative is strictly positive throughout, and $b_1^A \rightarrow 1$. As $\delta \rightarrow 1$ the condition recovers $(\mathbf{IC}_{b_1})$ and $b_1^A \approx 0.33$. $\square$

B Numerical verification and replication

All numerical values stated in the propositions are reproduced in the companion Jupyter notebook `Fixed_Budget_Appendix.ipynb`, which is organized section-by-section to mirror the paper and verifies every boxed numerical result. It is available at <https://github.com/sahmed95/fixed-budget-mh>.

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